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Regime Detection and Forecasting in Multi-Asset Markets: A Markov-Switching Bayesian SVAR Approach

Michael Ladaa

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Professor: Gabriele Deana

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ABSTRACT

Financial markets exhibit substantial variation in return dynamics, volatility, and cross-asset relationships over time. Traditional econometric models often assume constant parameters and are therefore unable to fully capture the regime-dependent nature of financial markets. This thesis investigates whether a Markov-Switching Heteroskedastic Bayesian Structural Vector Autoregressive framework can identify distinct market regimes, improve the understanding of cross-asset transmission mechanisms, and generate economically meaningful forecasts for portfolio allocation.

The empirical analysis is conducted using monthly data from February 1994 to April 2026 covering four major asset classes: equities (S&P 500), bonds (Bloomberg U.S. Aggregate Bond Index), the U.S. Dollar Index (DXY), and gold. The proposed framework combines Bayesian estimation, structural identification, and Markov-switching dynamics to allow both autoregressive relationships and volatility structures to vary across latent regimes. Model estimation is performed using a Gibbs sampler with embedded Metropolis-Hastings steps, while regime inference is obtained through Hamilton filtering and Kim smoothing.

The results provide strong evidence of regime-dependent behavior in financial markets. Four economically interpretable regimes are identified and classified as inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation. These regimes differ substantially in terms of conditional returns, volatility patterns, structural shock transmission, and persistence. Impulse response functions and forecast error variance decompositions reveal significant variation in the propagation and importance of structural shocks across market environments. Out-of-sample forecasting results show directional accuracies ranging from 63.9% to 75.0% across asset classes.

The forecasting framework is further evaluated through portfolio backtesting. A forecast-driven long-only strategy outperforms passive benchmark portfolios and a standard VAR-based benchmark, generating superior risk-adjusted returns during the evaluation period. The findings suggest that regime-switching Bayesian structural models provide a useful and interpretable framework for analyzing financial markets and can produce economically valuable information for portfolio allocation decisions.

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CHAPTER 1: INTRODUCTION

Financial markets are characterized by substantial time variation in volatility, correlations, and return dynamics. Periods of stable growth and low uncertainty are frequently interrupted by episodes of financial stress, inflationary shocks, recessions, or sudden changes in monetary policy expectations. During such periods, asset classes often behave fundamentally differently from their historical averages. Correlations between risky assets tend to increase, volatility rises sharply, and traditional diversification strategies may become substantially less effective. These recurrent shifts suggest that financial markets do not evolve according to a single stable data-generating process, but instead transition between distinct economic and financial environments.

The importance of regime dependence has become increasingly evident over recent decades. Major events such as the Dot-Com crisis, the Global Financial Crisis, the COVID-19 pandemic, and the recent inflationary environment following geopolitical tensions have demonstrated that the relationships between equities, bonds, currencies, and commodities can change considerably over time. In particular, the interaction between growth expectations, inflation dynamics, and monetary policy conditions plays a central role in shaping cross-asset return behaviour. Consequently, modelling frameworks that assume constant relationships across the full sample period may fail to adequately capture the nonlinear and state-dependent nature of financial markets.

Traditional econometric models, such as standard Vector Autoregressions (VARs), provide a flexible framework for analysing dynamic relationships among multiple time series. However, these models generally rely on the assumption that autoregressive coefficients and covariance structures remain constant over time. While this assumption simplifies estimation and interpretation, it is often restrictive in financial applications where market conditions evolve substantially across different economic states. In particular, constant-parameter models are unable to account for shifts in volatility regimes, changing transmission mechanisms, or state-dependent interactions between asset classes.

To address these limitations, the literature has increasingly focused on regime-switching approaches. Since the contribution of Hamilton [12], Markov-switching models have become an important tool for capturing nonlinear dynamics and latent state transitions in macroeconomic and financial time series. By allowing parameters to vary across unobserved regimes, these models provide a flexible framework capable of representing periods of calm market behaviour alongside turbulent crisis episodes. At the same time, Bayesian estimation methods have become increasingly important in high-dimensional macro-financial settings,

as they allow prior information and parameter uncertainty to be incorporated directly into the estimation process. Bayesian Vector Autoregressive models further address the overparameterization problems that frequently arise in multivariate systems.

Beyond capturing regime-dependent dynamics, understanding the structural transmission of shocks across asset classes remains a central objective in empirical finance. Reduced-form models describe correlations and predictive relationships, but they do not directly identify economically interpretable shocks. Structural Vector Autoregressive (SVAR) models therefore impose additional restrictions that allow innovations to be interpreted as distinct structural disturbances. However, conventional recursive identification strategies often rely heavily on ordering assumptions, which may be difficult to justify in complex financial systems. An alternative approach exploits heteroskedasticity across regimes as a source of identification. In this framework, changes in volatility across latent states provide additional information that helps recover the structural relationships underlying the observed data.

This thesis develops and applies a Markov-Switching Heteroskedastic Bayesian Structural Vector Autoregressive (MSH-BSVAR) framework to analyse regime-dependent dynamics across major financial asset classes. The proposed framework combines regime-switching behaviour, Bayesian estimation, and structural identification within a econometric setting. In particular, the model allows both the autoregressive dynamics and the covariance structure of financial returns to vary across latent economic regimes. The framework is designed to capture changing transmission mechanisms between equities, bonds, currencies, and gold, while simultaneously incorporating parameter uncertainty and probabilistic regime classification.

The empirical analysis focuses on four major asset classes represented by the S&P 500 Index, the Bloomberg U.S. Aggregate Bond Index, the U.S. Dollar Index, and gold prices. These assets capture key dimensions of global financial markets, including economic growth expectations, monetary policy conditions, currency movements, and inflation hedging behaviour. The analysis is conducted using monthly data spanning the period from February 1994 to April 2026. The model assumes four latent regimes, interpreted economically as inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation.

The objective of this thesis is threefold. First, the thesis aims to identify latent macro-financial regimes and analyse how asset return dynamics differ across these environments. Second, the thesis investigates whether structural transmission mechanisms and volatility structures vary substantially across regimes. Third, the thesis evaluates whether the resulting probabilistic forecasts contain economically meaningful predictive information that can be translated into practical portfolio allocation strategies.

The central research question of this thesis is therefore:

> *"Can a Markov-Switching Heteroskedastic Bayesian Structural Vector Autoregressive framework effectively identify regime-dependent dynamics across financial asset classes and generate economically meaningful forecasts that can be applied to portfolio allocation?"*

The contribution of this thesis lies in the integration of several themes of literature into a unified empirical framework. While previous research has extensively studied Markov-switching models, Bayesian VARs, and structural identification separately, comparatively limited attention has been given to combining regime-dependent structural identification, probabilistic forecasting, and cross-asset portfolio applications within a single Bayesian framework. In addition, the thesis contributes by implementing the complete estimation procedure in Python, including Gibbs sampling, Metropolis-Hastings updates, Hamilton filtering, Kim smoothing, regime permutation algorithms, structural impulse response analysis, and forecast-based portfolio backtesting.

Methodologically, the model is estimated using Bayesian Markov Chain Monte Carlo methods. A Gibbs sampler with embedded Metropolis-Hastings steps is employed to jointly estimate regime-dependent VAR coefficients, structural matrices, covariance structures, transition probabilities, and latent regime paths. Hamilton filtering and Kim smoothing are used to infer filtered and smoothed regime probabilities. Model evaluation is conducted using Bayesian diagnostics, impulse response functions, forecast error variance decompositions, out-of-sample forecasting metrics, and portfolio backtesting procedures.

The empirical results indicate substantial heterogeneity across regimes in both return dynamics and volatility structures. The estimated regimes align with economically interpretable macro-financial environments and display strong persistence over time. Moreover, the results suggest that structural transmission mechanisms differ considerably across regimes, particularly with respect to the interaction between equities and bonds. The forecasting exercise further demonstrates that the probabilistic framework contains economically meaningful predictive information, while the portfolio backtesting results indicate that forecast-based allocation strategies can outperform several passive benchmark approaches during the evaluation period.

The remainder of the thesis is structured as follows. Chapter 2 reviews the relevant literature on regime-switching models, Bayesian VARs, structural identification, and regime-based asset allocation. Chapter 3 presents the methodological framework, including the model specification, prior distributions, Bayesian estimation procedure, and forecasting methodology.

Chapter 4 describes the data and discusses the main statistical properties of the asset return series. Chapter 5 presents the empirical results, including regime interpretation, structural transmission analysis, forecasting performance, and portfolio backtesting results. Finally, Chapter 6 concludes and discusses limitations and potential directions for future research.

CHAPTER 2: LITERATURE REVIEW

2.1 Regime Dynamics in Financial Markets

It is well researched and established that financial markets exhibit distinct regimes characterized by changes in volatility, correlations, and macroeconomic conditions. Rather than following a single stable data-generating process, asset returns often display nonlinear dynamics that reflect shifts between different economic environments. This observation has motivated a substantial body of literature focusing on regime-dependent behaviour in financial markets.

An important advance in this field is provided by Hamilton [12]. In his framework he introduces Markov-switching models to capture fluctuations in business cycles. He theorises that the economy evolves through a set of latent states, allowing for discrete changes in model parameters across regimes. This approach provides a way to model nonlinearities and has since become a cornerstone in both macroeconomic and financial time series analysis.

Building on this foundation, Ang and Timmermann [3] provide extensive empirical evidence of regime changes in financial markets. Their work demonstrates that asset return distributions, volatilities, and correlations vary systematically across different states. In particular, financial markets are characterized by periods of low volatility and stable correlations, as well as turbulent phases marked by heightened uncertainty and stronger co-movement across asset classes.

Another important empirical finding supporting regime-based modelling is the presence of volatility clustering and correlation breakdowns during crises. Longin et al. [23] and Ang et al. [2] show that correlations between equity markets increase significantly during downturns, thereby reducing the benefits of diversification in such periods. This behaviour underscores the importance of allowing for regime-dependent covariance structures.

Various approaches have been proposed to model the dynamics of regime transitions. In the standard Markov-switching framework, transition probabilities are assumed to be constant over time, implying that regimes are recurring and can be revisited. Extensions of this framework introduce additional flexibility. For instance, Durland and McCurdy [10] propose duration-dependent transition probabilities, while Doan et al. [9] and Diebold et al. [6] develop models in which transition probabilities depend on observable state variables. Despite these extensions, the assumption of constant transition probabilities remains widely used due to its tractability and interpretability.

Further crucial modelling choices concerns the interpretation and the number of regimes. In

macro-financial applications, regimes are often linked to combinations of economic growth and inflation dynamics. Campbell et al. [4] emphasize the joint role of growth and inflation risks in asset pricing, providing a natural foundation for classifying economic environments. Building on this idea, this thesis adopts a four-regime framework consisting of inflationary growth, deflationary growth, inflationary stagnation, and deflationary stagnation.

Overall, the evidence of regime-dependent dynamics in financial markets provides a strong motivation for the use of latent state models. By allowing parameters to change across unobserved regimes, these models offer a flexible framework to capture nonlinearities, time-varying risk, and structural shifts in asset return behaviour.

2.2 Vector Autoregressions in Macroeconomics and Finance

In empirical macroeconomics and financial econometrics vector autoregressive models have become one of the central tools. Since the seminal contribution of Sims [26], VAR models have provided a flexible framework for analysing dynamic interactions among multiple time series without imposing the strong a priori restrictions that characterized earlier structural macro econometric models. The key idea is to treat all variables in the system as jointly endogenous and to explain their current values by their own lagged values and the lagged values of all other variables in the system. This makes VARs particularly suitable for studying macro-financial linkages, where asset returns, interest rates, exchange rates, volatility measures, and commodity prices may interact dynamically over time.

A reduced-form VAR of order p can be written as

$$Y_t = c + \sum_{i=1}^{p} \Phi_i Y_{t-i} + u_t, \qquad u_t \sim \mathcal{N}(0, \Omega), \tag{2.1}$$

where Y_t is a $K \times 1$ vector of endogenous variables, c is a vector of intercepts, Φ_i are coefficient matrices capturing lagged dynamic relationships, and u_t denotes the reduced-form innovation with covariance matrix Ω. This compact representation allows each variable to react not only to its own history, but also to the past evolution of all other variables in the system. As emphasized by Lütkepohl [1] [16], this makes VARs a natural framework for modelling multivariate time series, forecasting, and studying the propagation of shocks.

In macroeconomics, VAR models are widely used to analyse the transmission of monetary policy shocks, business cycle fluctuations, inflation dynamics, and interactions between real and financial variables. In finance, they provide a useful framework for examining return predictability, cross-asset spillovers, volatility transmission, and dynamic correlations between asset classes. Their appeal lies in the fact that they impose relatively little theoretical structure at the estimation stage, while still allowing researchers to recover rich dynamic patterns from the data. Stock et al. [28] summarize this usefulness by highlighting with a few limitations four main applications of VAR models: data description, forecasting, structural inference, and policy analysis.

Two analytical tools make VAR models especially relevant for this thesis: impulse response functions and forecast error variance decompositions. Impulse response functions trace the dynamic effect of an innovation in one variable on the future path of all variables in the system. Forecast error variance decompositions, in contrast, quantify the contribution of different shocks to the forecast uncertainty of each variable. These tools are central in structural VAR analysis because they transform estimated dynamic relationships into economically interpretable measures of shock transmission and risk decomposition [16]. In the context of financial markets, this is particularly valuable because it allows to study how shocks to one asset class propagate to others over time.

Despite their flexibility, standard VAR models rely on the assumption that the autoregressive coefficients and the covariance matrix of the innovations are constant over the full sample period. This assumption is often restrictive in financial applications. As discussed in the previous section, asset markets are characterized by changing volatility, time-varying correlations, and structural breaks between calm and crisis periods. A constant-parameter VAR is therefore unable to distinguish between different market environments unless these changes are explicitly incorporated into the model. This limitation is especially relevant when the objective is to analyse regime-dependent transmission mechanisms, since the effect of a shock may differ substantially between, for example, a low-volatility growth regime and a high-volatility stress regime.

Consequently, the standard VAR framework provides an important methodological benchmark but is insufficient for the purposes of this thesis. It captures dynamic interdependencies between asset classes, but it does not allow these dynamics or the covariance structure to change across latent economic states. This motivates the use of extensions that introduce both parameter uncertainty and regime dependence. Bayesian VAR models address the problem of overparameterization and estimation uncertainty through prior shrinkage, while Markov-switching VAR models allow the parameters of the system to vary across unobserved regimes.

2.3 Bayesian VAR Models

While vector autoregressive models provide a framework for capturing interactions among macroeconomic and financial variables, their implementation is often challenged by overparameterization. As the number of variables and lags increases, the number of parameters grows rapidly, leading to imprecise estimates and poor out-of-sample performance. This issue is particularly pronounced in financial applications, where datasets are often limited relative to the dimensionality of the model. To address these limitations, Bayesian methods have been introduced to incorporate prior information and regularize estimation [22]. The Bayesian VAR (BVAR) framework extends the classical VAR by treating model parameters as random variables and combining prior beliefs with the information contained in the data. Thus, inference is based on the posterior distribution

$$p(\Theta \mid Y) \propto p(Y \mid \Theta) \cdot p(\Theta), \tag{2.2}$$

where $p(Y \mid \Theta)$ denotes the likelihood function and $p(\Theta)$ represents the prior distribution over the parameter vector Θ. This approach allows for a coherent assessment of model uncertainty, which does not exist in classical VAR estimation. As emphasized by Sims and Zha [27], Bayesian methods provide a full multivariate predictive distribution rather than point forecasts alone, thereby offering a more comprehensive framework for inference and decision-making.

A key component of the BVAR approach is the use of shrinkage priors to mitigate overparameterisation. The most used example is the Minnesota prior introduced by Litterman [22]. In particular, the prior assumes that variables show persistence, implying that coefficients on own lags are likely to be close to unity, while coefficients on other variables and higher-order lags are shrunk towards zero. This structure effectively reduces estimation uncertainty by imposing economically meaningful restrictions without eliminating flexibility entirely. Early contributions by Doan, Litterman, and Sims [8] demonstrate that such priors can substantially improve forecasting performance in large VAR systems.

In addition to the Minnesota prior, a widely used specification in Bayesian VARs is the conjugate Normal-Inverse-Wishart prior. Here, the VAR coefficients are assumed to follow a multivariate normal distribution conditional on the covariance matrix, while the covariance matrix itself follows an inverse-Wishart distribution. Formally, this can be written as

$$\alpha \mid \Sigma \sim \mathcal{N}(\bar{\alpha}, \Sigma \otimes \bar{V}), \tag{2.3}$$

$$\Sigma \sim \mathcal{IW}(\bar{S}, \bar{\nu}), \tag{2.4}$$

where $\bar{S}$ and $\bar{\nu}$ denote the scale matrix and degrees of freedom, respectively. This property leads to tractable conditional posterior distributions, which significantly simplify estimation. As discussed by Koop and Korobilis [19], conjugate priors allow for efficient implementation of Bayesian VARs, particularly when combined with simulation-based methods.

Estimation in the Bayesian framework is typically carried out using Markov Chain Monte Carlo methods, as the joint posterior distribution is generally not available in closed form. Among these methods, the Gibbs sampler plays a central role. The Gibbs sampler iteratively draws from the conditional posterior distributions of parameter blocks, generating a sequence of dependent samples that converge to the joint posterior distribution. This approach is particularly effective in BVAR settings due to the availability of closed-form conditional distributions under conjugate priors. A comprehensive treatment of Gibbs sampling and its properties is provided by Geweke [11]. In more complex settings, where certain conditional posterior distributions are not available in closed form, the Gibbs sampler can be augmented with Metropolis-Hastings steps. The Metropolis-Hastings algorithm generates candidate draws from a proposal distribution and accepts them with a probability that ensures convergence to the target posterior distribution. This hybrid MCMC approach is widely used in Bayesian econometrics and allows for flexible modelling of parameters that do not admit conjugate updates [11].

2.4 Structural VAR Identification

The preceding sections established VAR models as a framework for modelling dynamic interactions among variables. However, for many empirical applications it is not sufficient to describe comovements or forecast future values. In order to interpret the effect of economically meaningful shocks, additional structure is required. While the residuals of a standard VAR capture unexpected movements in the variables, they are generally contemporaneously correlated among variables. Therefore, they cannot directly be interpreted as independent structural shocks. As emphasized by Sims [26], the reduced-form VAR alone does not uniquely determine the underlying structural relationships. Consequently, identification restrictions are required to recover economically interpretable shocks from the estimated VAR system.

The structural VAR representation introduces contemporaneous relationships between the variables through a structural impact matrix. A structural VAR of order p can be written as

$$AY_t = c + \sum_{i=1}^{p} B_i Y_{t-i} + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, \Sigma), \tag{2.5}$$

where A captures contemporaneous interactions between the variables and ε_t denotes the vector of structural shocks. These shocks are assumed to be mutually uncorrelated, such that Σ is typically diagonal. Premultiplying the structural system by A^{-1} yields the reduced-form VAR,

$$Y_t = \mu + \sum_{i=1}^{p} \Phi_i Y_{t-i} + u_t, \qquad u_t \sim \mathcal{N}(0, \Omega), \tag{2.6}$$

where the reduced-form covariance matrix is given by

$$\Omega = A^{-1}\Sigma(A^{-1})'. \tag{2.7}$$

This relationship illustrates the identification problem. Although Ω can be estimated from the data, it does not uniquely determine the structural matrix A and the structural shock variances contained in Σ. Different combinations of A and Σ may imply the same reduced-form covariance matrix and are therefore observationally equivalent. To obtain a unique structural interpretation, additional restrictions must be imposed [16].

A widely used approach is recursive identification, which imposes short-run restrictions on the contemporaneous relationships between variables. In this case, the structural matrix A is restricted to be lower triangular, corresponding to a Cholesky-type identification scheme. This restriction imposes an ordering of the variables: variables ordered earlier in the system may contemporaneously affect variables ordered later, while variables ordered later can affect earlier variables only with a lag. As discussed by Kilian and Lütkepohl [16], this approach provides a transparent and computationally convenient way to map reduced-form innovations into orthogonal structural shocks.

The main advantage of structural identification is that it allows the dynamic effects of shocks to be interpreted economically. Once structural shocks are identified, impulse response functions can be used to trace how a shock to one variable affects the entire system over time. Similarly, forecast error variance decompositions can quantify how much of the forecast uncertainty of each variable is attributable to each structural shock. These tools are therefore not merely statistical summaries, but provide an economically interpretable decomposition of the transmission of shocks across variables. In the context of this thesis, this is particularly relevant for analysing how shocks propagate across asset classes under different market conditions.

At the same time, recursive identification is not without limitations. Its interpretation depends on the ordering of variables, and alternative orderings may imply different structural impulse responses. Other identification strategies, such as sign restrictions or long-run restrictions, have therefore been proposed in the structural VAR literature. Nevertheless,

recursive identification remains widely used because it is transparent, easy to implement, and provides a clear mapping from statistical innovations to structural shocks [16].
The model used in this thesis builds on this structural VAR framework but extends it to a regime-dependent setting. Instead of assuming a single structural impact matrix for the entire sample, the contemporaneous relationships between variables are allowed to vary across latent regimes. Hence, for regime s, the structural system can be written as

$$A_s Y_t = c_s + \sum_{i=1}^{p} B_{i,s} Y_{t-i} + \varepsilon_{t,s}, \qquad \varepsilon_{t,s} \sim \mathcal{N}(0, \Sigma_s). \tag{2.8}$$

This specification allows both the transmission of shocks and the volatility of structural innovations to differ across market environments. Thus, the regime-specific structural matrices provide the basis for interpreting shock propagation in a framework that is consistent with the regime-dependent dynamics discussed earlier.

2.5 Markov-Switching Models

The preceding sections established that financial markets are characterised by recurring shifts in volatility, correlation structure, and macroeconomic conditions. Capturing these dynamics requires a modelling framework that accommodates discrete changes in the data-generating process rather than imposing parameter constancy throughout the sample. Markov-switching models, as originally proposed by Hamilton [12], provide precisely such a framework. Their central insight is that the economy may occupy one of a finite number of latent states, with each state governed by its own set of parameters and the transitions between states following a probabilistic law.
The latent state at time t is represented by a discrete random variable $S_t \in \{1, \ldots, M\}$, which is unobservable and assumed to evolve according to a first-order Markov chain. Under this assumption, the probability of transitioning from regime i to regime j depends only on the current state and not on the history of prior regimes. The transition structure is fully characterised by the matrix P, with elements

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij}, \qquad \sum_{j=1}^{M} p_{ij} = 1 \quad \forall\, i. \tag{2.9}$$

The diagonal entries p_{ii} govern regime persistence: values close to unity imply that a regime, once entered, tends to endure for an extended period, whereas lower values imply more frequent transitions. This feature enables the model to reproduce actual market cycles such as prolonged expansions, recessions, and bursts of financial volatility without imposing a

fixed periodicity on these phenomena.

Embedded within a vector autoregressive framework, the Markov-switching specification allows both the conditional mean dynamics and the innovation covariance structure to vary across states.

The resulting MS-VAR model takes the form

$$Y_t = \mu_{S_t} + \sum_{i=1}^{p} \Phi_{i,S_t} Y_{t-i} + u_t, \qquad u_t \mid S_t \sim \mathcal{N}(0, \Omega_{S_t}), \tag{2.10}$$

where the intercept vector μ_{S_t}, autoregressive coefficient matrices Φ_{i,S_t}, and innovation covariance matrix Ω_{S_t} are all indexed by the current underlying regime. This formulation thus permits regime-specific transmission dynamics alongside regime-specific second-moment structures. As documented by Longin and Solnik [23] and Ang and Chen [2], asset return correlations increase markedly during periods of market stress, a pattern that constant-parameter models are structurally unable to accommodate. Comprehensive treatments of MS-VAR models are provided by Krolzig [21] and Lanne, Lütkepohl, and Maciejowska [24].

Because S_t is latent, inference requires probabilistic techniques that assign, at each point in time, a likelihood to each possible regime. The Hamilton filter [12] accomplishes this through a forward-recursive algorithm that, at each date t, combines the predicted regime probabilities with the likelihood of the observed data to produce the filtered probabilities

$$\xi_{t|t} = \mathbb{P}(S_t = s \mid \mathcal{F}_t), \tag{2.11}$$

where $\mathcal{F}_t$ denotes the information set available at time t. These quantities represent the model's real-time assessment of the prevailing economic state and form the basis for parameter estimation via the Expectation-Maximisation algorithm.

Filtered probabilities, however, exploit only the information accumulated up to time t and are consequently less precise than estimates that draw on the complete sample. The Kim smoother [17] addresses this limitation by appending a backward recursion to the Hamilton filter, yielding the smoothed probabilities

$$\xi_{t|T} = \mathbb{P}(S_t = s \mid \mathcal{F}_T), \tag{2.12}$$

which condition on the full dataset. Relative to their filtered counterparts, smoothed probabilities exhibit markedly lower revision variance and provide a more precise ex-post classification of regimes, rendering them the preferred basis for empirical interpretation and historical attribution of economic states.

Within this thesis, the Markov-switching framework outlined above serves a dual purpose.

First, it provides the structural apparatus for representing regime-dependent dynamics across the four macroeconomic environments introduced in Section 2.1. Second, and more critically for identification, the heterogeneity in regime-specific covariance matrices Ω_{S_t} can be exploited to recover structural shocks. This identification strategy, originally proposed by Lanne et al.[24] and further developed by Herwartz and Lütkepohl [14], uses the cross-regime variation in second moments as a source of identifying information.

2.6 Identification via Heteroskedasticity

The mentioned structural identification problem arises because the reduced-form covariance matrix Ω alone is insufficient to recover the structural impact matrix A uniquely [26]. Recursive identification resolves this indeterminacy by imposing a triangular structure on A, an assumption that is computationally convenient, but whose validity depends entirely on the chosen variable ordering. Where economic theory offers limited guidance on contemporaneous restrictions, such assumptions may be difficult to justify [16]. An alternative identification strategy, which places fewer demands on a priori economic reasoning, exploits variation in the second-moment structure of the data itself [24] [25].
The central insight underlying identification via heteroskedasticity is that changes in the variance of structural shocks across regimes generate multiple observations of the same structural system under distinct covariance conditions. While the structural impact matrix A is assumed to be invariant across regimes, the diagonal covariance matrix of structural shocks Σ_{S_t} is not. Consequently, the regime-specific reduced-form covariance matrices

$$\Omega_{S_t} = A^{-1}\,\Sigma_{S_t}\,(A^{-1})', \tag{2.13}$$

constitute a system of matrix equations in the single unknown A. Provided that the matrices Σ_{S_t} differ sufficiently across regimes, this system is over-identified relative to the recursive case: each regime supplies an independent set of moment conditions, and their combination pins down A without requiring any element to be fixed at zero. Thus, each regime offers a distinct cross-sectional view of the same underlying structural relationships, and it is the diversity of these views that resolves the identification problem.

This identification strategy was formalised in the context of Markov-switching VAR models by Lanne, Lütkepohl, and Maciejowska [24], who establish the conditions under which heteroskedasticity achieves point identification of the structural parameters. A necessary condition is that the regime-specific variance matrices are mutually non-proportional, ensuring

that each regime conveys distinct second-moment information. This condition is empirically testable and, as documented in the regime dynamics literature reviewed in Section 2.1, is generally satisfied in financial data where volatility shifts between calm and stress periods are both pronounced and persistent. The approach has subsequently been extended to a Bayesian setting by Lütkepohl and Woźniak [25], who develop a posterior simulation framework that treats structural parameters, regime-specific variances, and latent state sequences as jointly unknown quantities.

Within the framework adopted in this thesis, the identification approach integrates naturally with the estimation strategy outlined in Section 2.3. The joint posterior over the structural matrix A, the regime-specific variance matrices Σ_s, the autoregressive coefficients, and the latent state sequence is not available in closed form and is therefore approximated via Markov Chain Monte Carlo methods. Specifically, parameters admitting tractable conditional posteriors under the Normal-Inverse-Wishart conjugate structure are sampled using the Gibbs sampler, while the structural parameters entering A non-linearly are updated via Metropolis-Hastings steps. This hybrid approach, discussed in detail by Geweke [11], propagates both parameter uncertainty and regime uncertainty through to all downstream quantities of interest, including impulse response functions and forecast error variance decompositions.

2.7 Regime-Based Asset Allocation

The design of regime-dependent dynamics in financial markets has important implications for portfolio construction. As discussed, asset return behaviour is not stable over time but instead varies systematically across different economic environments. These variations are not limited to changes in volatility, but also extend to correlation structures, asset returns and therefore to the effectiveness of diversification strategies. From an asset allocation perspective, this implies that portfolio performance is linked to the underlying market regime. A key pillar of modern portfolio theory is the advantage of diversification, which relies on weak correlations between asset classes. However, empirical evidence suggests that these correlations are themselves state dependent. Correlations between risky assets tend to increase significantly during periods of market stress, thereby reducing the effectiveness of diversification precisely in times of downturns. This phenomenon, which is also often referred to as correlation breakdown, is well documented in the literature. Longin and Solnik [23] show that international equity correlations rise sharply in extreme market conditions, while Ang and Chen [2] provide evidence of asymmetric correlations that strengthen during downturns. More broadly, Ang and Timmermann [3] demonstrate that both return distributions and cross-asset relationships vary systematically across regimes.

These findings challenge the validity of static asset allocation strategies that assume stable covariance structures and constant risk-return trade-offs. If the joint distribution of asset returns depends on the underlying regime, then portfolio weights that are optimal in one regime may be suboptimal or even highly exposed to risk in another. Consequently, asset allocation decisions should account for the possibility of regime shifts and adapt to changing market conditions.

Recent contributions in the asset management literature explicitly incorporate regime dependence into portfolio construction. Kinlaw, Kritzman, and Turkington [18] introduce the concept of co-occurrence, which captures the joint behaviour of assets across different economic states and highlights the importance of conditioning allocation decisions on regime-specific characteristics.

Similarly, Kritzman [20] emphasizes that regime ambiguity and uncertainty about the prevailing state should be incorporated into portfolio decisions, rather than relying on a single deterministic classification of market conditions. These approaches suggest that effective portfolio management requires not only identifying regimes but also accounting for the uncertainty associated with them.

The modelling framework developed in this thesis directly supports such regime-aware allocation strategies. By estimating latent regimes and their associated dynamics, the model provides probabilistic assessments of the current market environment as well as regime-dependent forecasts of asset returns and volatilities. This information can be used to adjust portfolio exposures in a systematic manner, thereby aligning investment decisions with the underlying economic state.

In this context, regime-dependent forecasts form the basis for translating econometric insights into actionable investment strategies. As will be discussed in the following section, these forecasts can be mapped into trading signals and evaluated within a backtesting framework.

2.8 Forecast-Based Portfolio Strategies

The developed framework provides a rich set of probabilistic forecasts. These can be used both for economic interpretation, as well as for practical decision-making. In particular, the joint estimation of regime probabilities, conditional return dynamics, and volatility structures enables the construction of forecasts of expected asset performance. However, to translate these econometric insights into actionable investment decisions, it is necessary to map model outputs into concrete portfolio strategies.

A natural approach is to base asset allocation decisions on the models forecasts of future

returns and risks. In a general setting, portfolio weights depend on expected returns, variances, and covariances of asset returns [5]. Within a regime-switching framework, these quantities become state-dependent, implying that optimal portfolio allocations vary across different economic environments. As a result, forecast-based allocation strategies can be interpreted as dynamic extensions of classical portfolio choice, where investment decisions adapt to changing market conditions and incorporate information about regime probabilities.

In practice, the implementation of such strategies often relies on simplified decision rules that translate forecasts into trading signals. A common approach is to construct sign-based or threshold-based signals [7], where predicted positive returns lead to long positions, while negative forecasts imply reduced exposure or short positions. These rules provide an easy way to convert complex forecasts into implementable portfolio decisions, while preserving the directional information contained in the model. Such signal-based approaches are widely used in empirical finance, as they allow for transparent evaluation and straightforward economic interpretation.

The empirical performance of forecast-based strategies is typically assessed within a backtesting framework. This involves evaluating the strategy on out-of-sample data, ensuring that all forecasts are generated using only information available at the time the decision is made. As emphasized by Diebold [7], this real-time perspective is crucial for avoiding look-ahead bias and obtaining an unbiased assessment of predictive performance. Similarly, Campbell, Lo, and MacKinlay [5] highlight the importance of rigorous evaluation methods in empirical asset pricing, including the use of realistic trading assumptions and appropriate performance metrics.

Within such a framework, portfolio performance is commonly evaluated using measures such as cumulative returns, volatility, Sharpe ratios, and drawdowns. These metrics provide a comprehensive view of both the return-generating capacity and the risk characteristics of the strategy. Importantly, they allow for a direct comparison between model-based strategies and benchmark portfolios, thereby assessing whether the additional model complexity translates into economically meaningful improvements in performance.

The modelling approach adopted in this thesis integrates into this forecast-based investment framework. By providing regime-dependent forecasts of asset returns and volatilities, the model generates the inputs required for signal construction and portfolio allocation. In particular, the probabilistic nature of the regime classification allows investment decisions to reflect uncertainty about the current economic state, rather than relying on deterministic regime assignments.

CHAPTER 3: METHODOLOGY

3.1 Overview of Methodology

This chapter presents the methodological foundation of the analysis, which is based on a Markov-Switching Heteroskedastic Bayesian Structural Vector Autoregressive model. The framework aims to capture regime-dependent dynamics in financial markets, allowing for both changes in structural relationships and volatility across distinct economic states. By combining a Markov-switching mechanism, Bayesian estimation, and structural identification, the model provides a coherent way to analyze market behavior, interpret shocks, and generate forecasts that can be used in practical asset allocation decisions, while maintaining explainability.

The methodology proceeds in several stages:

First, the model parameters are defined, including regime-specific autoregressive coefficients, covariance matrices, transition probabilities, and structural matrices. The model is then initialized using pre-specified restrictions and starting values to ensure identification and numerical stability. Here, Bayesian prior distributions such as Litterman shrinkage priors incorporate economic plausibility while allowing for uncertainty.

Estimation is then carried out using a Gibbs sampler, which iteratively draws model parameters and latent regimes from their conditional posterior distributions. The Markov process is estimated via Hamilton filtering and Kim smoothing, yielding both filtered and smoothed regime probabilities. After estimation, the model is structurally identified through regime-specific impact matrices that allow for changing contemporaneous relationships between the different asset classes. The resulting structural shocks provide an interpretable decomposition of risk and transmission dynamics.

To assess model performance, Bayesian diagnostics such as Marginal Data Density and Savage-Dickey Density Ratio are employed for model comparison and hypothesis testing. Further, the estimated posterior draws are analyzed to compute regime-dependent Impulse Response Functions, Forecast Error Variance Decompositions, and transition matrices. Posterior means and MCMC densities are evaluated to assess convergence and regime characteristics. A permutation and labeling algorithm ensures that regimes are consistently matched to their economic interpretation across runs.

Finally, the model is used for forecasting applications, transforming posterior predictions into binary trading signals that enable empirical evaluation and back testing of strategy performance. This bridges the methodological estimation with practical financial decision-making,

linking probabilistic regime forecasts to portfolio allocation.

3.2 Model Specification and Parameters

The objective of this models' framework is to capture regime-dependent dynamics and variances across asset classes, which are assumed to be unobserved and evolving over time. Unlike a standard VAR model, which assumes constant parameters throughout the sample, this model allows all model components to depend on a latent regime indicator. This assumption is a crucial part of this model, as financial markets can show different periods with alternating volatility and correlations between asset classes.
We define the vector of observed variables as follows:

$$Y_t = \begin{pmatrix} y_{1,t} & y_{2,t} & \dots & y_{K,t} \end{pmatrix}^\top \tag{3.1}$$

where Y_t contains the returns of K financial assets observed at time $t = 1, ..., T$. The assets include a set of different classes such as equities, bonds, commodities and currencies. This allows the model to capture the interactions across the different assets. Further, the economy is assumed to switch between a finite number of regimes:

$$S_t \in \{1, \dots, M\}, \tag{3.2}$$

where S_t is unobserved and follows a first order Markov process. Each regime is interpreted as a distinct economic or financial environment, including own distinct characteristics.

3.2.1 Structural VAR Representation

The structural VAR represents the core of the model. It allows contemporaneous interactions between variables to be interpreted economically.
For a given regime s, the structural form of a VAR of order p is specified as follows:

$$A_s Y_t = c_s + \sum_{i=1}^{p} B_{i,s} Y_{t-i} + \varepsilon_{t,s}, \qquad \varepsilon_{t,s} \sim \mathcal{N}(0, \Sigma_s) \tag{3.3}$$

Here, A_s represents the structural matrix, which captures the versatile relationships between the asset classes. By allowing A_s to vary across regimes the model permits the transmission of shocks to change between market regimes or different periods. The vector c_s contains regime-specific intercepts, while the matrices $B_{i,s}$ denote the lagged dynamics. Finally, the structural shocks $\varepsilon_{t,s}$ are assumed to have their mean at zero, while their variance is regime-

dependent.
To ensure interpretability, structural shocks are assumed to be orthogonal within each regime. This assumption implies a diagonal covariance matrix:

$$\Sigma_s = \text{diag}\left(\sigma_{1,s}^2, \ldots, \sigma_{K,s}^2\right) \tag{3.4}$$

Essentially, this ensures that each shock can be interpreted as a distinct economic disturbance and it facilitates further impulse response analysis.
Throughout this thesis the structural form above will be strictly for interpretation purposes. For the act of estimation the reduced form is used out of convenience.
The reduced form is obtained by multiplying the structural equation by the inverse of A_s:

$$Y_t = \mu_s + \sum_{i=1}^{p} \Phi_{i,s} Y_{t-i} + u_{t,s}, \qquad u_{t,s} \sim \mathcal{N}(0, \Omega_s) \tag{3.5}$$

Here, we obtain the reduced-form parameters, which we will be referring to from now on. The parameters are simply functions of the structural parameters and are denoted by:

$$\mu_s = A_s^{-1} c_s, \tag{3.6}$$

$$\Phi_{i,s} = A_s^{-1} B_{i,s}, \tag{3.7}$$

$$\Omega_s = A_s^{-1} \Sigma_s (A_s^{-1})^\top \tag{3.8}$$

In this representation, the error term $u_{t,s}$ captures regime-dependent volatility and correlation patterns. This is a key feature of the heteroskedastic specification. Further, the reduced form provides the basis for likelihood evaluation, filtering, and Bayesian estimation, which will be done in the following chapters.

3.2.2 Regime-Dependent Parameters

All the relevant parameters of the model are allowed to vary across regimes. For regime s, the parameter set is defined as follows:

$$\Theta_s = \{A_s,\, c_s,\, \{B_{i,s}\}_{i=1}^{p},\, \Sigma_s\} \tag{3.9}$$

This specification implies that both the conditional mean dynamics and the conditional variance-covariance structure differ across the distinct regimes. As a result, the model can capture different phenomena such as changing correlations between two asset classes or shifts in the persistence of asset returns.

3.2.3 Markov Regime Dynamics

The regime process S_t is modeled as a first order Markov chain, which implies that the probability of being in a certain regime solely depends on the previous periods regime. Thus, we have dynamic transition probabilities that are captured in the formula below:

$$\mathbb{P}(S_t = j \mid S_{t-1} = i) = p_{ij}, \qquad \sum_{j=1}^{M} p_{ij} = 1 \tag{3.10}$$

These probabilities are summarized in the transition matrix:

$$P = \begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1M} \\ p_{21} & p_{22} & \cdots & p_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ p_{M1} & p_{M2} & \cdots & p_{MM} \end{pmatrix} \tag{3.11}$$

High values along the diagonal indicate persistent regimes, while larger off-diagonal elements imply frequent switches between two regimes. The transition matrix, hence establishes a crucial part of this work and provides direct information about the stability and expected duration of economic phases.
Combining the observation equation with the regime dynamics yields a state-space representation, where the observed data depends on an unobserved regime state. Conditional on the regime, the observation density is given by:

$$Y_t \mid S_t = s \sim \mathcal{N}\left(\mu_s + \sum_{i=1}^{p} \Phi_{i,s} Y_{t-i}, \Omega_s\right) \tag{3.12}$$

This formulation provides the foundation for Hamilton filtering, Kim smoothing, and Bayesian inference, which are discussed in subsequent sections.

3.3 Restrictions and Initialization

Setting appropriate restrictions and initialization values is an essential part of regime-switching models with structural identification. When done right, these can ensure both economic interpretability and numerical stability. The role of the restrictions is to ensure that the parameters of the Structural VAR are uniquely determined, while the initialization establishes a stable convergence in the Bayesian sampling algorithm. [26] [27]

Issues in these two steps are particularly relevant in Markov Switching Bayesian VARs, where the parameter space is high dimensional and includes latent regime indicators, such as in our underlying model. In these settings, weak identification or arbitrary starting values may lead to poor performance of the Markov Chain Monte Carlo algorithm or to economically meaningless regime allocations. [11] [17] In this framework the restrictions and initializations serve two crucial purposes: They ensure that the structural shocks can be interpreted economically in a more facilitated manner and they provide a stable starting point for posterior simulation, thus facilitating the convergence of the Gibbs sampler.

3.3.1 Identification Restrictions

To identify structural shocks the reduced form of the VAR is not sufficient as multiple structural representations can be observationally equivalent. By imposing identification restrictions on the structural impact matrices A_s, it is possible to recover economically meaningful shocks. [26]

In each regime s, the matrix A_s is assumed to be invertible and subject to restrictions. The restrictions employed in this model are a recursive triangular structure. This recursive identification corresponds to a lower-triangular structural matrix, analogous to a Cholesky-based ordering which imposes a causal ordering among the variables. Thus, the recursive restrictions provide the identification of the structural shocks, while the regime-dependent volatility structure allows the relationships between asset classes to vary across different market environments.

Therefore, the structural matrix takes the following form:

$$A_s = \begin{pmatrix} 1 & 0 & \dots & 0 \\ a_{21,s} & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ a_{K1,s} & a_{K2,s} & \dots & 1 \end{pmatrix}. \tag{3.13}$$

The assumption of this matrix structure is that variables ordered earlier can contemporaneously affect all the variables ordered later. Recursive identifications such as this one are widely used in the SVAR literature due to their simplicity and interpretability. [26] [16]

Another crucial aspect of the model is that it allows these structural matrices to be regime specific. Thus, it enables the relationships between the asset classes to change across different market phases. This aligns with our assumption where shock effects often differ strongly between different economic regimes.

Moreover, two regularity conditions are set to ensure a well defined model.

First, the VAR dynamics within each regime are assumed to be stable. This means that the eigenvalues of the companion matrix lie inside of the unit circle. With this condition explosive behavior of shocks is avoided, while also ensuring finite unconditional moments. [16]
Second, the regime-specific covariance matrices of the structural shock have to be positive definite, ensuring that the likelihood function is properly defined.
These two constraints are enforced through the prior specification and rejection sampling. Thus, by discarding invalid parameter draws during posterior simulation.

3.3.2 Initialization of Model Parameters

Given the complexity of the model, careful initialization is required to ensure efficient posterior sampling.
This initialization process consists of multiple steps.
The first step includes estimating the standard VAR without regime allocation using ordinary least squares. Here, the resulting coefficient estimates and residual covariance matrix are our baseline values for the parameters.
Thereafter, we classify the observation into regimes by using an expectation-maximization procedure based on the differences in volatility. This allows us to obtain initial regime probabilities that reflect broad differences in market conditions.
Finally, regime-specific VAR parameters are initialized by estimating separate models conditional on the preliminary regime assignments.
This multi stage initialization aims to improve numerical stability and convergence behavior. [21]

3.4 Prior Distributions

Most importantly all prior distributions are specified symmetrically across the different regimes. This means that initially no regime is favored over another in terms of volatility, dynamics or persistence, in order to avoid asymmetries in the regime classification.
Therefore, differences between regimes arise solely from the data after simulation trough the posterior distributions.

3.4.1 VAR Coefficients

The coefficients of the reduced form VAR are assigned using a Normal-Inverse Wishart prior, which is the standard conjugate prior for multivariate Gaussian regression models.

Conditional on the regime s, the prior takes the following form:

$$\alpha_s \mid \Sigma_s \sim \mathcal{N}\left(\bar{\alpha}, \Sigma_s \otimes \bar{V}\right), \tag{3.14}$$

$$\Sigma_s \sim \mathcal{IW}\left(\bar{S}, \bar{\nu}\right). \tag{3.15}$$

Where α_s stacks the intercepts and lag coefficients of the VAR in regime s.
To improve shrinkage, the prior mean and variance of the VAR coefficients are constructed using a Minnesota / Litterman prior. This means that the coefficients on own lags are centered towards unity, while coefficients on other variables and higher lags are shrunk towards zero. The choice of this prior reflects the belief that financial time series are persistent and that cross-variable effects are typically weaker.
The degree of the shrinkage factor λ is controlled by hyperparameters that determine overall lag decay and cross asset weighting. These hyperparameters are chosen by a grid search and updated during estimation.

3.4.2 Structural Parameters

For the structural matrices A_s the free elements are collected in a parameter vector a_s, which is assigned a multivariate normal prior:

$$a_s \sim \mathcal{N}\left(\bar{a}, \bar{V}_a\right). \tag{3.16}$$

This prior is centered at zero. This shows that there is a lack of strong beliefs about cross asset effects. Given that we do not have strong prior information about cross asset effects, the prior is centered at zero. Thus, we do not assume any systematic effects before observing the data.
The variance which acts as a shrinkage parameter is chosen sufficiently large so that the data can dominate posterior inference.
Moreover, the structural parameters are sampled using a Metropolis Hastings steps with the Gibbs sampler, as their conditional posterior does not have a closed form solution. Nevertheless, the normal prior ensures a consistent posterior density and numerical stability.

3.4.3 Shock Variances

The variances of the structural shocks are regime-specific and captured by the diagonal elements of the covariance matrix. An inverse-gamma prior is assigned to each structural

parameter and takes the following form.

$$\sigma_{j,s}^2 \sim \mathcal{IG}\left(\bar{\alpha}_\sigma, \bar{\beta}_\sigma\right). \tag{3.17}$$

This ensures that variances provide sufficient flexibility to capture the heteroskedasticity between different market regimes. In this framework, the distribution is conjugate, allowing the parameters to be updated directly within the Gibbs Sampler.

3.4.4 Transition Matrix

For the transition matrix P, each row of P is assigned a Dirichlet prior, which ensures that the transition probabilities are positive and sum to one.

$$(p_{i1}, \ldots, p_{iM}) \sim \mathcal{D}\left(\delta_{i1}, \ldots, \delta_{iM}\right). \tag{3.18}$$

Furthermore, the hyperparameters of the Dirichlet prior are chosen to reflect persistence in regimes. This implies the assumption that regimes are long lasting. While it is still possible to transition to another regime, it is most probable to remain in the previous one. This behavior is consistent with empirical observations of economic and financial cycles. [3]

3.5 Bayesian Estimation via Gibbs Sampling

The estimation strategy of the model relies on Markov Chain Monte Carlo (MCMC) methods. Due to the presence of regime-dependent parameters, latent indicators, and structural identification, the joint posterior distribution of the said unknown quantities does not have a closed form solution. Thus, estimation is conducted using a block-wise Gibbs sampler, which iteratively draws from the conditional posterior distributions. [27] [19] The Gibbs Sampler has a block structure that contains the set of all unknown and latent states. Here, each iteration is an interplay of sampling and updating these values. This is done until convergence is reached.

The Gibbs Sampler begins by sampling the VAR coefficients and the shock variances. Due to the Normal-Inverse Wishart priors that were defined earlier, the coefficients for each regime obtain a Gaussian conditional posterior, conditional on the regime sequence and the covariance matrix. Therefore, for each regime s, we obtain the following:

$$\alpha_s \mid \Sigma_s, S_{1:T}, Y \sim \mathcal{N}\left(\bar{\alpha}_s^*, \bar{V}_s^*\right). \tag{3.19}$$

In order to allow both parameter and regime uncertainty to be reflected in the estimation, the posterior moments depend on the observations which were associated with regime s, weighted by the corresponding probability of being in that regime.
The structural shock variances also depend on the given regime and are heteroskedastic across the different market phases. Conditional on the VAR coefficients and regime assignments, each variance parameter is sampled independently from an inverse-gamma distribution. For a variable j in regime s, we obtain:

$$\sigma_{j,s}^2 \mid \alpha_s, S_{1:T}, Y \sim \mathcal{IG}\left(\alpha_\sigma^*, \beta_\sigma^*\right). \tag{3.20}$$

This conjugate update of the prior ensures positive values for variances, while allowing flexible volatility levels across regimes. [27] [11]
Using the sampled values the latent regime sequence $S_{1:T}$ gets updated using a Forward-Backward algorithm, which combines Hamilton filtering and Kim smoothing. [12] [17] The smoothed regime probabilities incorporate information from the entire sample and are then computed as followed:

$$p(S_t = s \mid Y_{1:T}), \tag{3.21}$$

Based on these probabilities, a regime path is drawn conditional on the current parameter values.
The last step needed for inference, is the sampling of the Transition Matrix and the Structural Parameters. Conditional on the obtained regime sequence the transition probabilities are updated using their Dirichlet posterior distribution. For each row i of the transition matrix we update by:

$$(p_{i1}, \ldots, p_{iM}) \mid S_{1:T} \sim \mathcal{D}\left(\delta_{i1}^*, \ldots, \delta_{iM}^*\right). \tag{3.22}$$

This revision captures the persistence of regimes and their switching behavior while also ensuring that their transition probabilities remain valid. [12] Finally, the structural parameters contained in the structural matrices A_s do not have closed form conditional posterior. Therefore, they are updated using a Metropolis-Hasting step within the Gibbs Sampler. Thus, draws are generated from a multivariate normal distribution, while acceptance is determined by the standard Metropolis-Hastings probability. The acceptance probability depends on the ratio of the likelihood and prior evaluated at the proposed and current parameter values. [27] [11]
Having obtained all the necessary values, the posterior draws from the Gibbs sampler are then stored for inference. These draws will be the basis for future analysis, such as impulse response functions, variance decompositions, and forecasting. To ensure convergence it is

assessed using standard MCMC diagnostics such as trace plots and posterior moments. By using these conjugate updates and block-wise sampling we aim to improve the mixing while also reducing the autocorrelation in the Markov Chain. [11]

3.5.1 Filtering and Smoothing of Regimes

As already discussed the regime variable S_t is not directly observable and must therefore be inferred from the data. In our framework this inference is carried out by a combination of Hamilton Filtering and Kim Smoothing. The combination of the two methods provides a probabilistic classification of market states and give us a starting point for posterior sampling. Moreover, this probabilistic classification reflects uncertainty about regime timing as it assigns probabilities for each regime at each given time instead of assigning observations deterministically.
This approach allows us to provide the probabilities for obtaining the samples needed for a full regime path $S_{1:T}$. This is done with a Forward-Backward algorithm that ensures regime assignments that are consistent with both the likelihood and the transistion structure.

3.5.1.1 Hamilton Filter

The first step in regime inference is the Hamilton filter. [12] This filter simply computes the probability of being in regime s at time t, conditional on the information that is available for that given time period.

$$p(S_t = s \mid Y_{1:t}) \tag{3.23}$$

Essentially, the Hamilton filter represents the real time belief about the current regime based solely by recursively combining the transition probabilities with the regime specific likelihood of each observation.

3.5.1.2 Kim Smoothing

Unlike the Hamilton Filter, the Kim Smoothing incorporates the full sample, instead of just using past information. The aim of the Smoother is it to refine the regime probabilities by conditioning on all observations. [17]

$$p(S_t = s \mid Y_{1:T}) \tag{3.24}$$

This backward recursion adjusts the already filtered probabilities using future information and is used to produce smoother and more stable regime estimates.

3.6 Model Comparison: Marginal Data Density and Savage-Dickey Density Ratio

In this thesis, model comparison is conducted using the Marginal Data Density (MDD) and the Savage-Dickey Density Ratio (SDDR). These tools adhere with the Bayesian framework and help evaluate model fit and testing hypotheses. [15]

3.6.1 Marginal Data Density (MDD)

The Marginal Data Density is used to measure the overall fit of a model by integrating the likelihood over the prior distribution of the parameters. It is defined as follows:

$$p(Y) = \int p(Y \mid \Theta), p(\Theta), d\Theta \tag{3.25}$$

where Θ denotes the complete parameter vector of the model.
The MDD includes a regularization term, as it captures both the model fit and its complexity. This means that, unlike the classical likelihood, it penalizes over-parametrization. This is done given that more complex models would spread the prior mass over a larger parameter space, which would lead to favor complex models with redundant variables. [15]
To ensure numerical stability, the MDD is approximated using importance sampling. Thus, drawing from a proposal distribution, which is constructed from posterior simulations. Moreover, this approach allows us to compare models across different regime specifications and lag lengths. [11]
The evaluation of the score is straightforward, as higher values of the MDD indicate a stronger empirical support for a model.

3.6.2 Savage-Dickey Density Ratio (SDDR)

The Savage-Dickey Density Ratio is used to evaluate parameter restrictions within a given model. The SDDR provides a direct Bayesian measure of evidence in favor of a restricted hypothesis. [29]
For a restriction of the form $\theta = \theta_0$, the SDDR is defined as follows:

$$\frac{p(\theta = \theta_0 \mid Y)}{p(\theta = \theta_0)} \tag{3.26}$$

Thus, the SDDR compares the posterior density at the restricted value to the corresponding prior density. If the posterior density at θ_0 exceeds the prior density, the data provides

evidence in favor of the restriction and vice versa.
Therefore, the SDDR serves as a Bayesian analogue to hypothesis testing, as we are directly quantifying the evidence trough density ratios.

3.7 Permutation and Economic Regime Labeling

Markov-switching models do not impose an intrinsic ordering on regimes. The numerical labels assigned during estimation are arbitrary and may vary across different runs of the Gibbs sampler. This phenomenon, commonly referred to as label switching, can complicate interpretation and comparison across specifications.
To ensure consistent economic interpretation, a deterministic permutation procedure is applied after estimation. The regimes are reordered according to economically meaningful criteria derived from posterior moments. In particular, regimes are classified based on the conditional means and volatility patterns of key asset classes.
The ordering reflects economically interpretable states such as inflationary/deflationary growth or inflationary/deflationary stagnation. By imposing this economically motivated sorting rule, regime labels remain stable across estimation runs and allow coherent comparison of impulse responses, forecast behavior, and transition dynamics.

3.8 Impulse Response Functions and Forecast Error Variance Decomposition

The structural form of the model allows the dynamic effects of economically interpretable shocks to be analyzed. In this thesis this will be done through impulse response functions (IRF) and forecast error variance decompositions (FEVDs). The regime specific nature of the this model implies that both IRF and FEVD can differ substantially across different regimes.

3.8.1 Impulse Response Functions

Impulse Response Functions represent the reaction of the different asset classes to a one-unit structural shock. Formally, the moving average representation of a regime dependent VAR can be written as follows:

$$Y_t = \sum_{h=0}^{\infty} \Psi_{h,s} \varepsilon_{t-h,s}, \tag{3.27}$$

where $\Psi_{h,s}$ denotes the impulse response matrix at horizon h in regime s.
These matrices are obtained by recursively computing the reduced-form parameters. Fundamentally, they describe how a structural shock today affects each variable over future periods. Here the affect can differ from one economic state to the other, as the IRFs are regime specific.
Posterior distributions of the impulse responses are obtained by computing responses for each Gibbs draw and summarizing them using posterior means and credible intervals. [16]

3.8.2 Forecast Error Variance Decomposition

The Forecast Error Variance Decomposition quantifies the relative importance for each structural shock in explaining forecast uncertainty. [16] For horizon H, the forecast error variance of variable i om regime s can be expressed as follows:

$$\mathrm{Var}\left(Y_{i,t+H} - \hat{Y}_{i,t+H}\right) = \sum_{h=0}^{H-1} \sum_{j=1}^{K} \left(\Psi_{h,s}^{(i,j)}\right)^2 \sigma_{j,s}^2, \tag{3.28}$$

where $\Psi_{h,s}^{(i,j)}$ denotes the response of variable i to shock j at horizon h.
The FEVD is then computed as the proportion of this variable which can be attributed to each structural shock. Thus, it decomposes the forecast uncertainty into economically interpretable components.
Also here the FEVD is computed for each posterior draw and summarized using posterior moments.

3.9 Forecasting and Binary Forecast Evaluation

The model is ultimately used for predictive purposes. In order to ensure parameter and regime uncertainty, forecasting is also conducted within the Bayesian framework by integrating over posterior uncertainty. In order to efficiently seize the models outputs, the resulting out-of-sample predictions are transformed into trading signals, which are then evaluated in a portfolio backtesting. [13] [7].

3.9.1 Out-of-Sample Signal Generation and Timing

Out-of-sample (OOS) forecasts are converted into discrete trading signals at a monthly rebalancing frequency. The implementation uses month-end timestamps and constructs signals

based on the sign of predicted returns. For asset i at time t, the binary signal is defined as:

$$\text{Signal}_{i,t} = \begin{cases} 1 & \text{if } \widehat{y}_{i,t} > \tau, \\ 0 & \text{otherwise,} \end{cases} \tag{3.29}$$

where $\widehat{y}_{i,t}$ denotes the point forecast and τ is a threshold (set to 0 in the baseline specification). For the long-short strategy, the code additionally constructs ternary signals:

$$\text{Signal}_{i,t}^{LS} = \begin{cases} +1 & \text{if } \widehat{y}_{i,t} > \tau, \\ -1 & \text{if } \widehat{y}_{i,t} < -\tau, \\ 0 & \text{otherwise.} \end{cases} \tag{3.30}$$

A key aspect for unbiased evaluation is the treatment of timing. Signals are assumed to be published at month-end t and are applied to portfolio weights in month $t+1$. This shift avoids look-ahead bias and ensures that forecasts are evaluated in a realistic decision-making setup [7].

3.9.2 Portfolio Construction

Based on the signals, two portfolio strategies are constructed: a long-only strategy and a long-short strategy. Both strategies use an equal-weight rule, which ensures transparency and avoids introducing additional optimization assumptions beyond the model-based signal.

3.9.2.1 Long-Only Forecast Portfolio

In the long-only portfolio, all assets with a positive signal are included with equal weights, while all other assets are assigned weight zero. Let N_t^+ denote the number of assets with $\text{Signal}_{i,t} = 1$. Then portfolio weights are defined as:

$$w_{i,t}^{LO} = \begin{cases} \frac{1}{N_t^+} & \text{if Signal}_{i,t} = 1 \text{ and } N_t^+ \geq N_{\min}, \\ 0 & \text{otherwise,} \end{cases} \tag{3.31}$$

where $N_{\min}$ denotes the minimum number of long positions required. If fewer than $N_{\min}$ assets have a positive signal, the strategy remains fully in cash for that month.

3.9.2.2 Long-Short Forecast Portfolio

The long-short portfolio assigns equal absolute weight to all non-zero signals. Let N_t^A denote the number of active positions at time t. Then:

$$w_{i,t}^{LS} = \begin{cases} \frac{1}{N_t^A} & \text{if Signal}_{i,t}^{LS} = +1, \\ -\frac{1}{N_t^A} & \text{if Signal}_{i,t}^{LS} = -1, \\ 0 & \text{otherwise.} \end{cases} \tag{3.32}$$

This construction normalizes the gross exposure such that $\sum_i |w_{i,t}^{LS}| = 1$, and any remaining weight is treated as cash. If no active signals exist, the strategy stays in cash.

3.9.3 Transaction Costs, Turnover, and Cash Yield

To reflect implementability, the backtest accounts for transaction costs and the return on uninvested capital. First, portfolio turnover is measured as the monthly L^1 change in weights:

$$\text{Turnover}_t = \sum_{i=1}^{K} |w_{i,t} - w_{i,t-1}| \,. \tag{3.33}$$

Transaction costs are given in basis points and are modeled as a proportional round-trip cost applied to the traded notional:

$$TC_t = \text{Turnover}_t \cdot \frac{c}{10{,}000}, \tag{3.34}$$

where c denotes the round-trip cost in basis points.
Second, the strategy can hold cash whenever the portfolio is not fully invested for the long only, or when gross exposure is below one for the long-short. Cash earns a constant annualized rate r_f, converted to a monthly yield:

$$r_{f,m} = (1 + r_f)^{1/12} - 1. \tag{3.35}$$

Hence, net portfolio returns are computed as the gross portfolio return plus the cash component minus transaction costs.

3.9.4 Benchmarking and Economic Evaluation

To assess whether the proposed regime-dependent forecasting framework provides value beyond simpler investment approaches, the forecast-based strategies are compared against sev-

eral benchmark portfolios.
The first benchmark is an equal-weight buy-and-hold portfolio that allocates identical weights to all asset classes and maintains these allocations throughout the evaluation period. This benchmark represents a simple diversification strategy that does not rely on forecasting information and serves as a reference point for passive asset allocation.
The second benchmark is a forecast-driven VAR strategy. In contrast to the proposed framework, this benchmark relies on a standard rolling Vector Autoregression estimated on monthly asset returns. At each rebalancing date, the VAR model generates one-step-ahead return forecasts using only information available at that point in time. Portfolio weights are then assigned proportionally to positive return forecasts, while assets with non-positive forecasts receive zero weight. Consequently, the benchmark follows the same long-only portfolio construction logic as the proposed strategy, but excludes regime-switching dynamics and regime-dependent volatility structures. This comparison therefore aims to show the contribution of latent regime information and structural regime dependence.
Performance is evaluated using standard portfolio metrics, including cumulative performance, annualized volatility, maximum drawdown, and Sharpe ratios, consistent with standard practice in empirical asset allocation and backtesting. [5]

3.9.5 Random Buy-and-Hold Comparison

In addition to fixed benchmarks, the code implements a distributional comparison based on random buy-and-hold portfolios. Specifically, a large number of random long-only portfolios is simulated by drawing static weights from a Dirichlet distribution and holding them constant over the backtest window. The resulting distribution of terminal wealth provides a robustness check as it assesses whether the forecast-driven strategy performs unusually well (or poorly) relative to a broad class of naive buy-and-hold allocations.

CHAPTER 4: DATA

4.1 Data Description

The empirical analysis is based on a set of monthly financial time series covering the period from February 1994 to April 2026. The data is sampled at a monthly frequency using end-of-month observations, which ensures consistency with the forecasting framework and the subsequent portfolio rebalancing strategy.

The dataset comprises four asset classes representing different segments of the global financial market. Specifically, the analysis includes an equity index, a bond index, a currency index, and a commodity proxy. The selected variables are the S&P 500 Index (sp500), the Bloomberg U.S. Aggregate Bond Index (LUATTRUU Index), the U.S. Dollar Index (DXY Index), and the gold price (XAU Curncy). These assets capture key dimensions of financial markets, including growth, interest rate exposure, currency dynamics, and inflation hedging. The data up to July 2025 is obtained from Bloomberg, ensuring high-quality and widely used financial market data. Observations beyond this point are extended using data retrieved from Yahoo Finance. This combination allows for a consistent and up-to-date sample while maintaining compatibility with the initial data source. All series are aligned and cleaned prior to analysis.

4.1.1 Selected Asset Classes

The choice of asset classes is motivated by their distinct economic roles and their relevance for regime-dependent analysis. The S&P 500 Index serves as a proxy for equity markets and captures the performance of large U.S. corporations. It is typically associated with economic growth and is sensitive to changes in risk sentiment.

The Bloomberg U.S. Aggregate Bond Index represents fixed income markets and reflects interest rate dynamics and monetary policy conditions. Bonds are generally considered lower-risk assets and often act as a stabilizing component in diversified portfolios.

The U.S. Dollar Index (DXY) measures the value of the U.S. dollar relative to a basket of major currencies. It captures global capital flows, monetary policy differentials, and safe-haven dynamics, particularly during periods of financial stress.

Gold (XAU) is included as a representative commodity and is widely regarded as a store of value and an inflation hedge. It often exhibits different behavior compared to traditional financial assets, especially during periods of uncertainty or inflationary pressure.

Together, these four asset classes provide a broad representation of financial markets and allow the model to capture cross-asset interactions and regime-dependent dynamics.

4.1.2 Data Transformation

For the empirical analysis, the price series are transformed into continuously compounded returns. Specifically, the logarithmic return of asset i at time t is defined as

$$r_{i,t} = \log\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{4.1}$$

where $P_{i,t}$ denotes the price level of asset i at time t.
This transformation is standard in financial econometrics, as it stabilizes the variance of the series and ensures approximate stationarity, which is required for the VAR framework. Moreover, log-returns are additive over time and facilitate the interpretation of cumulative performance. No further filtering or data adjustments are applied.

4.2 Descriptive Statistics

Table 4.1 reports summary statistics of the monthly log-returns for all asset classes. The results highlight substantial differences in return characteristics across assets.
Equities and gold exhibit the highest average returns, with annualized means of approximately 8.1% and 7.7%, respectively. In contrast, bond returns are considerably lower, reflecting their role as lower-risk investments, while the U.S. dollar index displays a near-zero average return over the sample period.
In terms of volatility, equities and gold again stand out, with annualized standard deviations of approximately 15.8% and 15.7%, respectively. Bonds are substantially less volatile, with an annualized standard deviation of around 4.7%, confirming their stabilizing role. The U.S. dollar index exhibits intermediate volatility.
The distributional properties further indicate deviations from normality, particularly for equities, which display negative skewness and excess kurtosis. This suggests the presence of downside risk and extreme observations, which are typical features of financial return series.

Table 4.1: Descriptive Statistics of Monthly Log-Returns

Asset	Mean	Std	Skewness	Kurtosis	Mean (ann.)	Std (ann.)
S&P 500	0.0068	0.0456	-0.88	2.18	0.0814	0.1581
Bonds	0.0033	0.0135	-0.05	1.31	0.0394	0.0466
DXY	0.0001	0.0222	0.03	0.53	0.0018	0.0768
Gold	0.0064	0.0453	-0.08	1.07	0.0773	0.1568

4.3 Time Series Properties

Figure 4.1 displays the evolution of the asset price indices over time. A clear upward trend is observed for equities and gold, reflecting long-term growth and inflationary dynamics. In contrast, bond prices evolve more smoothly, while the U.S. dollar index exhibits relatively stable behavior without a pronounced trend.

Figure 4.2 presents the corresponding twelve month rolling volatility. The series reveal pronounced volatility clustering, with periods of relatively calm fluctuations alternating with episodes of heightened market stress. In particular, large changes are visible during major crisis periods, such as the global financial crisis and the COVID-19 pandemic.

These patterns indicate that both the mean and the variance of asset returns vary over time, providing empirical support for the use of regime-switching models. The observed clustering of volatility and the presence of extreme observations further motivate the heteroskedastic specification adopted in this thesis.

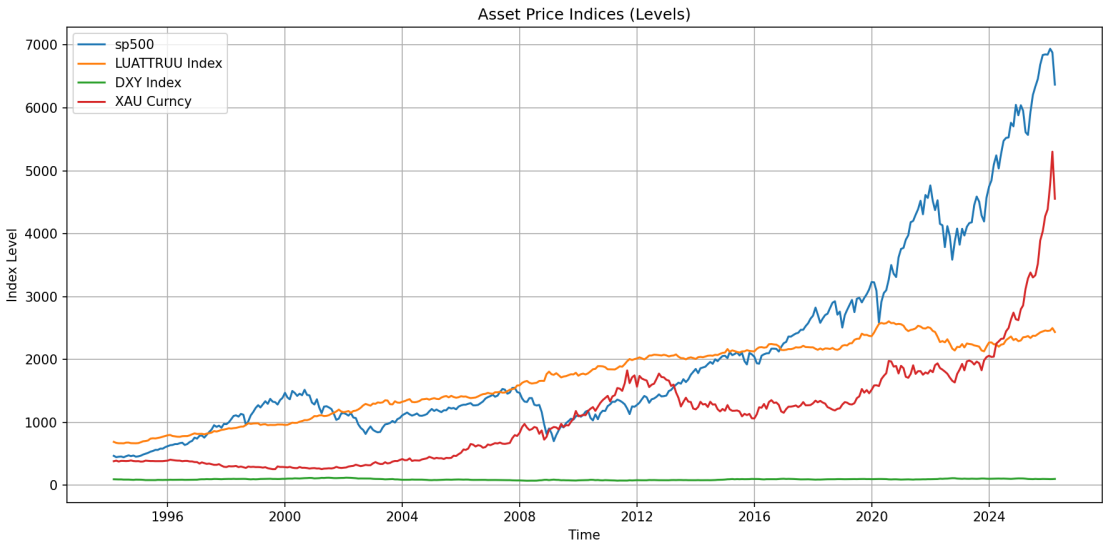


Figure 4.1: Asset Price Indices (Levels)

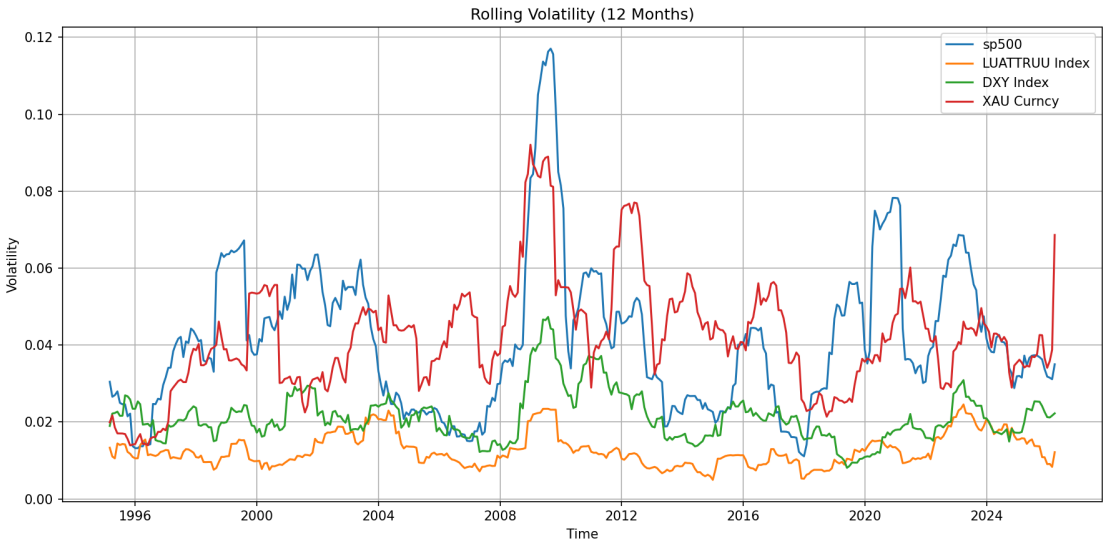


Figure 4.2: Monthly Log Returns

4.4 Implications for the Model

The characteristics of the data provide strong support for the modelling approach adopted in this thesis. The presence of time-varying volatility, volatility clustering, and differing risk-return profiles across asset classes suggests that a constant-parameter model would be insufficient to capture the underlying dynamics.

In particular, the coexistence of calm periods and turbulent episodes indicates the presence of latent regimes, while the heterogeneity in asset behavior motivates a multivariate framework capable of capturing cross-asset interactions. These features align naturally with the Markov-switching heteroskedastic Bayesian VAR model introduced in the previous chapter, which explicitly allows for regime-dependent dynamics in both the conditional mean and variance.

CHAPTER 5: RESULTS

5.1 Model Setup

The baseline specification of the used model assumes $M = 4$ latent regimes and a lag length of $p = 4$. The choice of four regimes is motivated by the macro-financial interpretation of economic environments as combinations of growth and inflation dynamics, as discussed in Section 2.1. Specifically, the regimes are interpreted as inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation. This classification provides an economically meaningful structure while allowing sufficient flexibility to capture heterogeneity in asset return dynamics. The lag length of four corresponds to a quarterly structure in monthly data and enables the model to capture short- to medium-term dependencies without introducing excessive parameterization. Further the choice of lag has been empirically confirmed by running a grid search.

Estimation is conducted using a Gibbs sampler with embedded Metropolis-Hastings steps for the structural parameters. A total of 5,000 posterior draws are generated, of which the first 2,000 iterations are discarded as burn-in. The remaining draws are used for posterior inference. These values reflect a trade-off between computational feasibility and estimation precision, given the high dimensionality of the parameter space and the presence of regime-switching dynamics. Convergence of the Markov chain is assessed using standard diagnostics, including trace plots and posterior stability checks.

Prior distributions are specified symmetrically across regimes to avoid introducing ex-ante biases in the regime classification. The VAR coefficients are assigned a Normal-Inverse Wishart prior with Litterman-type shrinkage, reflecting the persistence typically observed in financial time series. Structural parameters are assigned diffuse normal priors, while regime-specific variances follow inverse-gamma distributions. Transition probabilities are modeled using Dirichlet priors that favor persistent regimes, consistent with empirical evidence on financial market cycles.

Following estimation, a deterministic permutation procedure is applied to ensure consistent economic labeling of regimes. Since the ordering of regimes in Markov-switching models is inherently arbitrary, regimes are reordered based on posterior conditional mean returns of key asset classes, particularly equities and bonds. This step ensures that regimes are consistently interpreted across estimation runs.

For forecasting evaluation, the sample is split into an in-sample estimation period and a 36-month out-of-sample window. Forecasts are generated using posterior predictive distri-

butions, and their performance is evaluated based on binary accuracy. In addition, forecasts are used to construct trading signals for portfolio backtesting. The forecast horizon is set to 20 months for the analysis of impulse responses and variance decompositions, while shorter horizons are used for out-of-sample evaluation.

5.2 Regime Interpretation

This section interprets the models estimated regimes based on their conditional mean returns, volatility patterns, and historical regime probabilities. Since the model is estimated on financial return data, the identified regimes should not be interpreted as directly observed macroeconomic states, but rather as latent financial environments inferred from the joint behaviour of equities, bonds, the U.S. dollar, and gold.

The interpretation of the regimes is primarily based on the interaction between equities and bonds, as these two asset classes contain the clearest information about the underlying growth and inflation environment. In particular, the sign combination of conditional equity and bond returns forms the basis for the economic classification into inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation. This classification is implemented through a permutation procedure explained below, which ensures that regime labels remain economically consistent across estimation runs.

5.2.1 Conditional Mean Returns Across Regimes

Figure 5.1 presents the standardized conditional mean returns for each asset class and regime. The values are expressed as z-scores and therefore measure how strongly the regime-specific conditional mean deviates from the unconditional mean return of the respective asset class. Positive values indicate above-average expected returns, while negative values indicate below-average expected returns. The regime classification is based primarily on the interaction between equities and bonds. The permutation procedure assigns economic labels according to the relative sign and strength of these two asset classes. Intuitively, positive equity returns indicate favorable growth conditions, while bond performance provides information about inflationary pressure and interest-rate dynamics.

Inflationary growth is therefore characterized by positive equity returns combined with negative bond returns. This pattern is visible in the first regime, where the S&P 500 exhibits a strongly positive standardized mean of 0.76, while bonds display a negative value of -0.82. Such an environment is economically plausible in periods of strong nominal growth and elevated inflation, where equities benefit from improving earnings expectations while rising

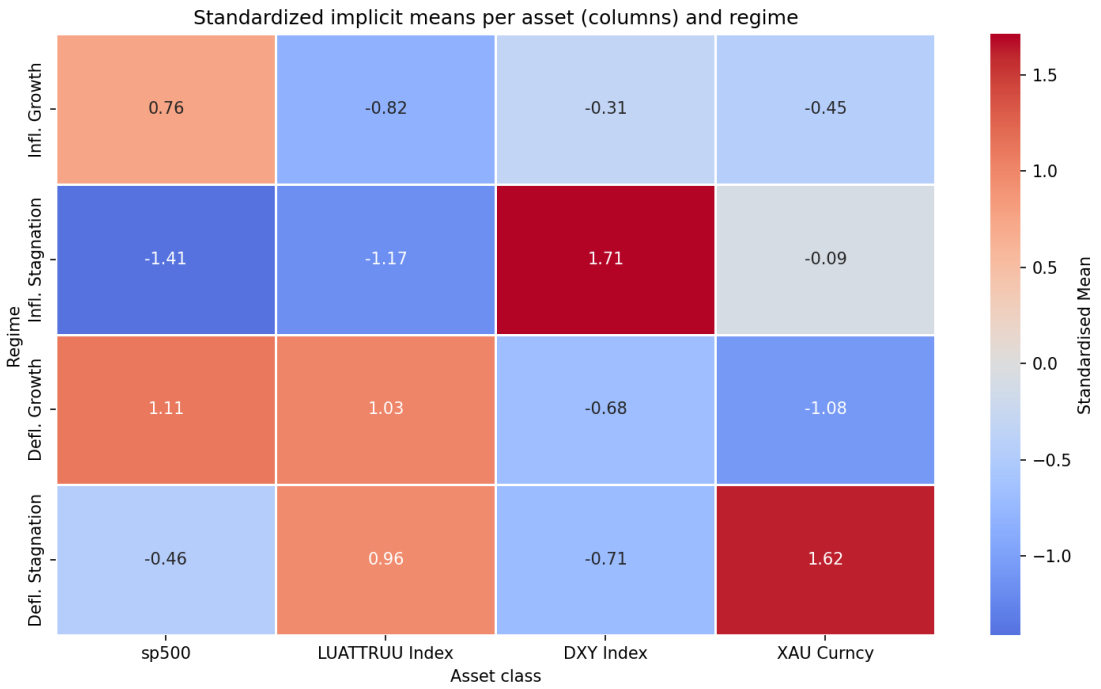


Figure 5.1: Standardized Conditional Mean Returns by Asset and Regime

interest rates negatively affect fixed-income markets.

Inflationary stagnation is identified by simultaneously weak equity and bond performance. In the second regime, equities show the most negative conditional mean across all regimes with a value of -1.41, while bonds also perform substantially below average at -1.17. This configuration is consistent with stagflationary environments in which weak growth expectations coexist with persistent inflationary pressure. Interestingly, the U.S. dollar exhibits the strongest positive conditional mean across all assets and regimes with a value of 1.71. This indicates strong demand for dollar exposure during periods of economic uncertainty and restrictive monetary policy.

Deflationary growth is associated with simultaneously positive equity and bond returns. In the third regime, the S&P 500 and bonds record strongly positive values of 1.11 and 1.03, respectively. This combination is consistent with stable growth combined with subdued inflationary pressure. Such environments support both corporate profitability and fixed-income valuations, thereby producing favorable conditions for traditional diversified portfolios. Gold and the U.S. dollar both exhibit below-average returns in this regime, indicating lower demand for hedging assets.

Finally, deflationary stagnation is characterized by weak equity performance combined with positive bond returns. In the fourth regime, equities remain below average at -0.46, while bonds exhibit a positive value of 0.96. At the same time, gold records the strongest positive

value within this regime at 1.62. This pattern is economically consistent with recessionary or crisis-like environments in which growth weakens, inflation declines, and investors seek defensive assets such as bonds and gold.

Overall, the conditional mean estimates strongly support the economic interpretation imposed by the permutation procedure. The regimes differ substantially in their cross-asset return behaviour and closely align with theoretical expectations regarding the interaction between growth, inflation, monetary conditions, and investor risk sentiment.

5.2.2 Regime-Dependent Volatility

While the conditional means determine the economic classification of regimes, the volatility structure provides additional information about the risk environment associated with each regime. Figure 5.2 reports standardized volatility estimates for each asset and regime. Similar to the conditional mean analysis, the values are expressed as deviations from the unconditional average volatility in standard deviations. From a theoretical perspective, higher

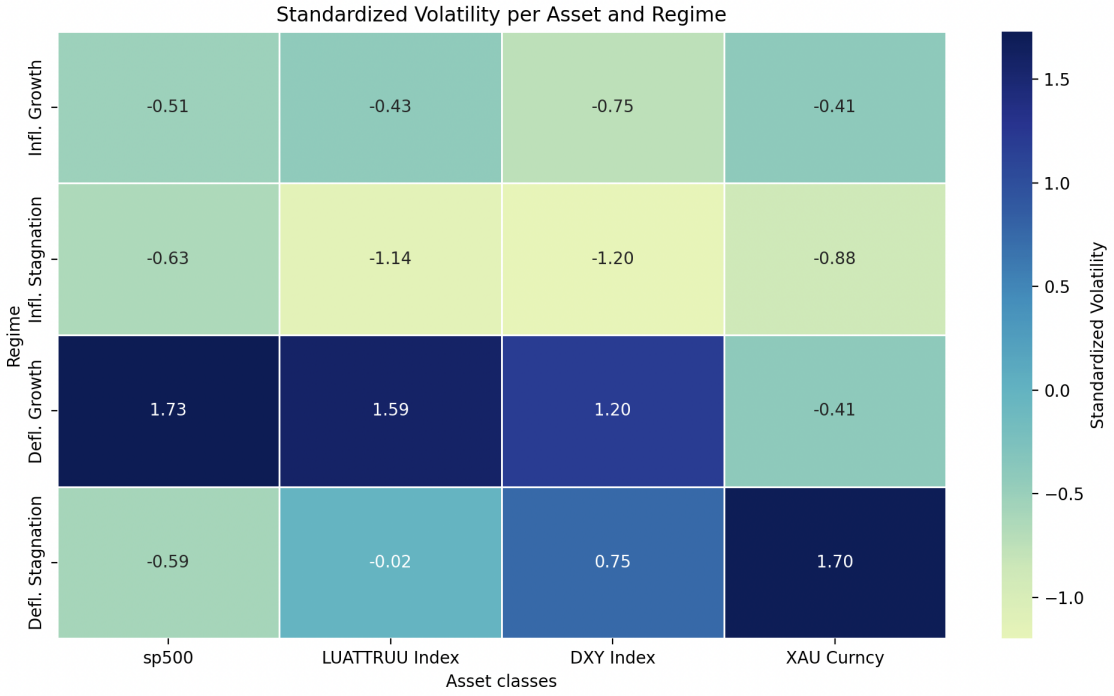


Figure 5.2: Standardized Volatility by Asset and Regime

volatility is generally expected during economically adverse periods. Financial crises, recessions, inflation shocks, and periods of uncertainty tend to increase disagreement among market participants and amplify reallocations across asset classes. This typically results in elevated volatility, stronger co-movement between risky assets, and increased safe-haven de-

mand.

The results indicate that volatility differs substantially across the identified regimes. Inflationary growth exhibits relatively low volatility across all asset classes. Equity volatility is moderately below average at -0.51, while bond, dollar, and gold volatility are also negative. This suggests that the inflationary growth regime reflects a relatively stable expansionary environment rather than a turbulent inflation shock.

Inflationary stagnation also exhibits below-average volatility across most assets. This result is particularly interesting because it suggests that unfavorable return conditions do not necessarily coincide with extreme market turbulence. Instead, the regime appears to capture a persistently weak environment characterized by low returns rather than crisis-like volatility. This distinction between poor performance and high volatility is important for interpreting the identified states.

The highest volatility regime is deflationary growth. Equities, bonds, and the U.S. dollar display strongly elevated volatility values of 1.73, 1.59, and 1.20, respectively. Although this initially appears counterintuitive for a growth regime, the result likely reflects transitional recovery phases following crises or major macroeconomic adjustments. During such periods, markets may exhibit strong positive returns while uncertainty remains elevated due to changing growth expectations, monetary-policy shifts, or financial-market reallocation.

Deflationary stagnation is primarily characterized by extremely high gold volatility, which reaches a standardized value of 1.70, alongside elevated dollar volatility. This pattern is highly consistent with flight-to-safety dynamics during periods of economic weakness or uncertainty. In such environments, investors frequently shift aggressively into defensive assets, thereby increasing volatility in gold and currency markets.

More generally, the volatility results highlight an important distinction between growth and stagnation regimes. Growth regimes are primarily differentiated by the behaviour of risky assets and the interaction between equities and bonds, whereas stagnation regimes are more strongly associated with defensive positioning and safe-haven demand. Moreover, the results indicate that volatility does not increase uniformly across all adverse environments. Instead, the type of macroeconomic stress determines which asset classes experience elevated uncertainty.

5.2.3 Historical Regime Dynamics and Current Regime

Figure 5.3 presents the smoothed regime probabilities over the full sample period. These probabilities are conditioned on the complete information set and therefore provide an ex-post classification of historical market environments. The regime probabilities exhibit strong

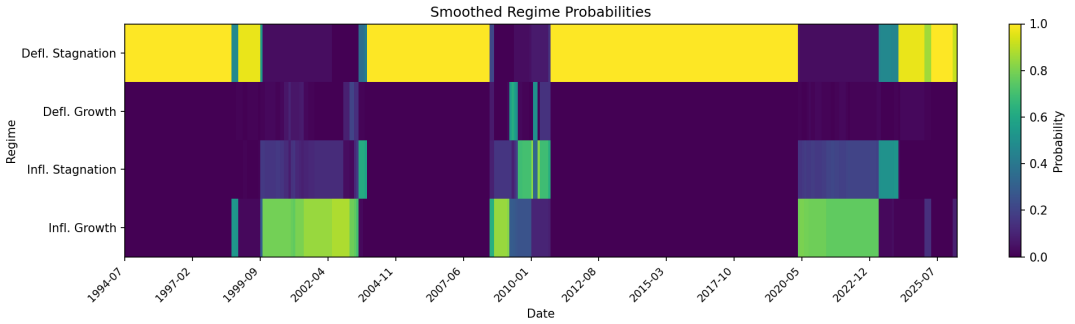


Figure 5.3: Smoothed Regime Probabilities

persistence, indicating that the model identifies prolonged market environments rather than short-lived fluctuations. This persistence is consistent with the estimated transition dynamics and with the broader literature on financial regimes.

Several major macroeconomic episodes are reflected clearly in the regime path. The late 1990s and early 2000s are primarily associated with inflationary growth and inflationary stagnation, corresponding to the transition from the technology boom to the subsequent market correction. The period surrounding the global financial crisis and the European sovereign debt crisis is predominantly classified as deflationary stagnation, which is economically intuitive given the collapse in growth expectations, aggressive monetary easing, and strong demand for defensive assets.

The years following the financial crisis are also dominated by deflationary stagnation probabilities. This classification is consistent with the prolonged low-growth and low-inflation environment that characterized the post-2008 period. During these years, global monetary policy remained highly accommodative, interest rates remained exceptionally low, and investors increasingly sought defensive assets.

The COVID-19 crisis again coincides with elevated deflationary stagnation probabilities, reflecting the severe contraction in economic activity and the heightened uncertainty during the pandemic period. The subsequent recovery phase beginning in 2020 temporarily increased inflationary growth probabilities, which is consistent with strong fiscal stimulus, reopening effects, and rising inflation pressure.

More recently, the model indicates a transition toward inflationary stagnation. Figure 5.4 shows that the most recent observation is classified as inflationary stagnation with a probability of approximately 68.7%.

This classification is economically plausible given the current macroeconomic environment. The inflationary stagnation regime is characterized by weak equity and bond performance, strong U.S. dollar behaviour, and relatively contained but persistent volatility. This resem-

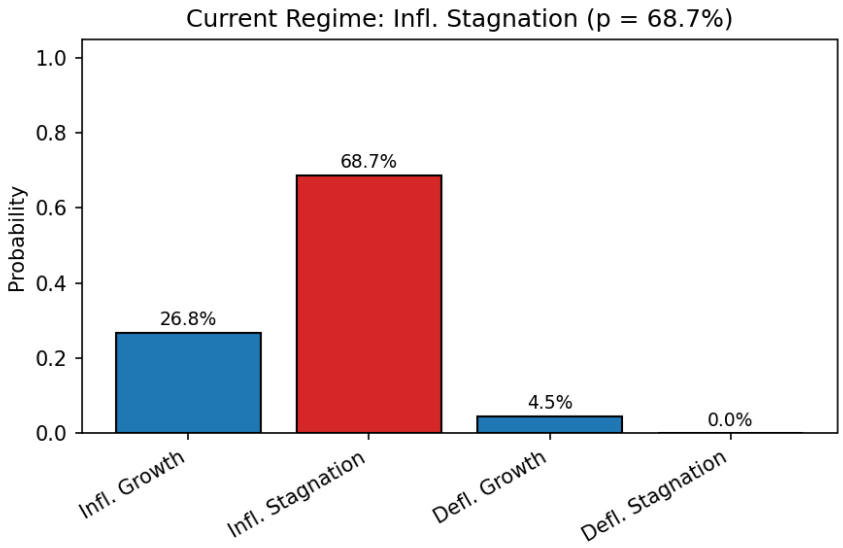


Figure 5.4: Current Regime Probabilities

bles an environment in which growth expectations weaken while inflation remains sufficiently elevated to prevent a rapid easing of monetary policy.

Recent macroeconomic developments and geopolitical tensions support this interpretation. Since 2022, global financial markets have been characterized by restrictive monetary policy, elevated geopolitical uncertainty, weaker global growth expectations, and inflation remaining above central-bank targets. At the same time, higher interest rates have weighed on both equity valuations and fixed-income markets, while the U.S. dollar has weakened compared to other major currencies. Ongoing conflicts, particularly in the Middle East, are currently reducing growth expectations due to rising commodity prices, while simultaneously increasing inflationary pressures and expectations of higher interest rates. These characteristics align closely with the estimated properties of the inflationary stagnation regime.

5.3 Transition Dynamics

The transition matrix in this section provides additional insight into the persistence and switching behaviour of the estimated regimes. While the previous section focused on the economic interpretation of regime-specific means and volatilities, the transition probabilities describe how likely the model is to remain in a given regime or transition into another regime from one month to the next.

In Markov-switching models, regime persistence is reflected in relatively large diagonal elements of the transition matrix. The diagonal probabilities represent the probability of

remaining in the same regime in the subsequent period. Persistent regimes are a central feature of the Markov-switching framework introduced by Hamilton [12], as business-cycle and financial-market environments are generally expected to last for several periods rather than switching randomly from month to month.

Figure 5.5 presents the estimated transition probabilities. The rows correspond to the current regime, while the columns represent the probability of transitioning into another regime in the following month. The estimated transition matrix exhibits a pronounced diagonal

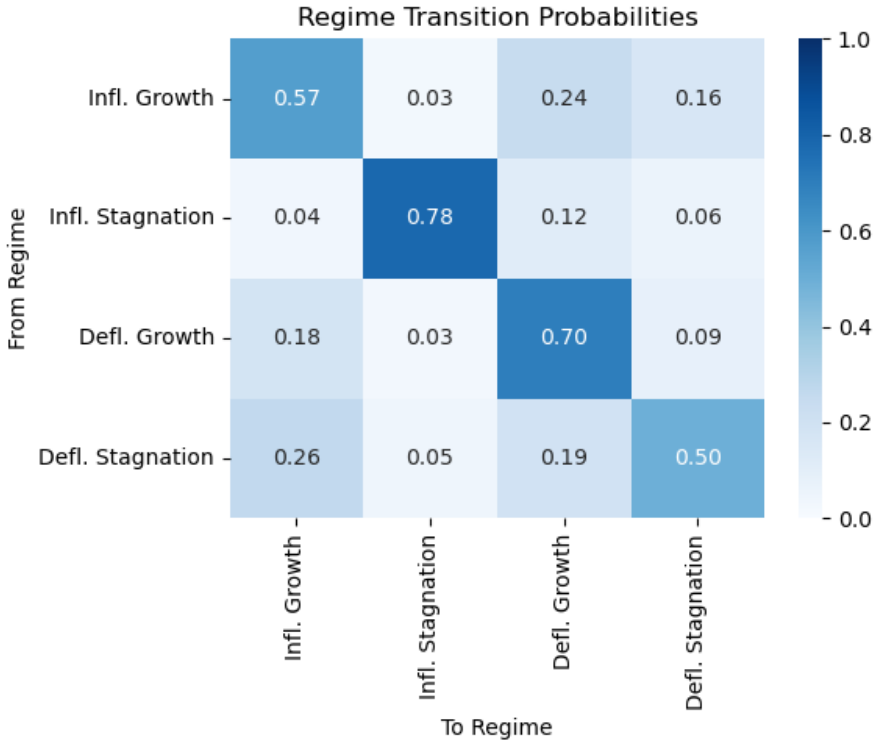


Figure 5.5: Estimated Regime Transition Probabilities

structure, indicating substantial regime persistence across all four states. The highest persistence is observed for the inflationary stagnation regime, which remains in the same state with a probability of 0.78. Deflationary growth also exhibits strong persistence with a diagonal probability of 0.70, followed by inflationary growth with 0.57 and deflationary stagnation with 0.50.

The expected duration of a regime can be approximated by

$$\mathbb{E}(D_i) = \frac{1}{1 - p_{ii}}, \tag{5.1}$$

where p_{ii} denotes the probability of remaining in regime i. Based on the estimated transition probabilities, the implied expected durations are approximately 2.33 months for inflationary growth, 4.55 months for inflationary stagnation, 3.33 months for deflationary growth, and 2.00 months for deflationary stagnation.

The inflationary stagnation regime therefore appears to be the most persistent state in the model. This result is economically plausible, as periods characterized by elevated inflation, restrictive monetary policy, and weak growth expectations often persist over extended periods. Inflationary pressure and monetary-policy tightening generally evolve gradually and are not resolved within only a few months. Consequently, once the economy enters such an environment, financial markets may remain trapped in a prolonged period of weak equity and bond performance combined with persistent dollar strength.

Deflationary growth also exhibits substantial persistence. This is consistent with stable low-inflation expansionary environments, which are typically associated with accommodative financial conditions, improving growth expectations, and relatively stable market dynamics. Such environments can persist for several quarters before inflationary pressures begin to emerge.

Inflationary growth is moderately persistent with a probability of 0.57. This suggests that positive nominal growth environments tend to last, but remain more vulnerable to transitions into other states compared to inflationary stagnation or deflationary growth. In particular, inflationary growth most frequently transitions into deflationary growth with a probability of 0.24. Economically, this may reflect situations in which inflationary pressure gradually subsides while growth remains positive, thereby producing a more favorable environment for both equities and bonds.

Deflationary stagnation exhibits the lowest persistence among the four regimes. Although the regime still remains in the same state with a probability of 0.50, it transitions relatively frequently toward inflationary growth with a probability of 0.26. This pattern is economically intuitive, as severe recessionary or crisis-like periods are often followed by strong policy responses, recoveries in growth expectations, and subsequent rebounds in risky assets.

The off-diagonal probabilities further provide insight into the dynamics of regime transitions. Inflationary stagnation exhibits a very low probability of switching directly into inflationary growth (0.04), indicating that markets rarely move immediately from a stagflationary environment into a broad expansionary phase. Instead, transitions are more likely to occur gradually through intermediate states such as deflationary growth. Similarly, deflationary growth has only a small probability of transitioning directly into inflationary stagnation (0.03), suggesting that abrupt transitions from stable low-inflation growth to stagflationary environments are relatively uncommon in the estimated model.

5.4 Structural Transmission Results

The regime-dependent structure of the model allows the contemporaneous relationships between asset classes to vary across different economic environments. This section analyses these transmission mechanisms using the estimated structural matrices, impulse response functions, and forecast error variance decompositions for each of the four regimes.

5.4.1 Structural Matrices

Figure 5.6 reports the posterior mean structural matrices for the four estimated regimes. The matrices are normalized such that the diagonal elements are equal to one. This normalization follows directly from the recursive identification strategy and implies that the diagonal entries capture the own structural shocks of each variable.

The lower-triangular structure reflects the recursive ordering imposed on the system. Variables ordered earlier in the system may contemporaneously affect variables ordered later, while the reverse effect occurs only with a lag. Consequently, the off-diagonal elements measure the contemporaneous transmission of shocks across asset classes.

Several economically meaningful differences emerge across regimes. In all four regimes, shocks originating from the equity market show substantial contemporaneous effects on bonds, the U.S. dollar and Gold. However, the magnitude of these effects differs considerably across market environments.

The strongest contemporaneous transmission from equities to bonds is observed in the inflationary growth regime, where the corresponding coefficient reaches approximately 0.77. Similarly, the contemporaneous impact of equity shocks on the U.S. dollar is particularly pronounced in this regime, with a coefficient close to 0.94. This suggests that during inflationary expansion phases, movements in risky assets strongly influence broader financial conditions and currency markets.

In contrast, the inflationary stagnation regime exhibits substantially weaker cross-asset transmission effects. The contemporaneous relationship between equities and bonds falls to approximately 0.04, indicating that asset classes move more independently during periods of stagflationary stress. This is economically plausible, as periods of inflationary stagnation are often characterized by conflicting macroeconomic signals and fragmented market dynamics.

The deflationary growth and deflationary stagnation regimes display intermediate transmission patterns. In particular, the deflationary stagnation regime shows a relatively strong contemporaneous effect from equities to bonds (0.67), while gold remains comparatively weakly connected to the remaining financial system. This supports the interpretation of

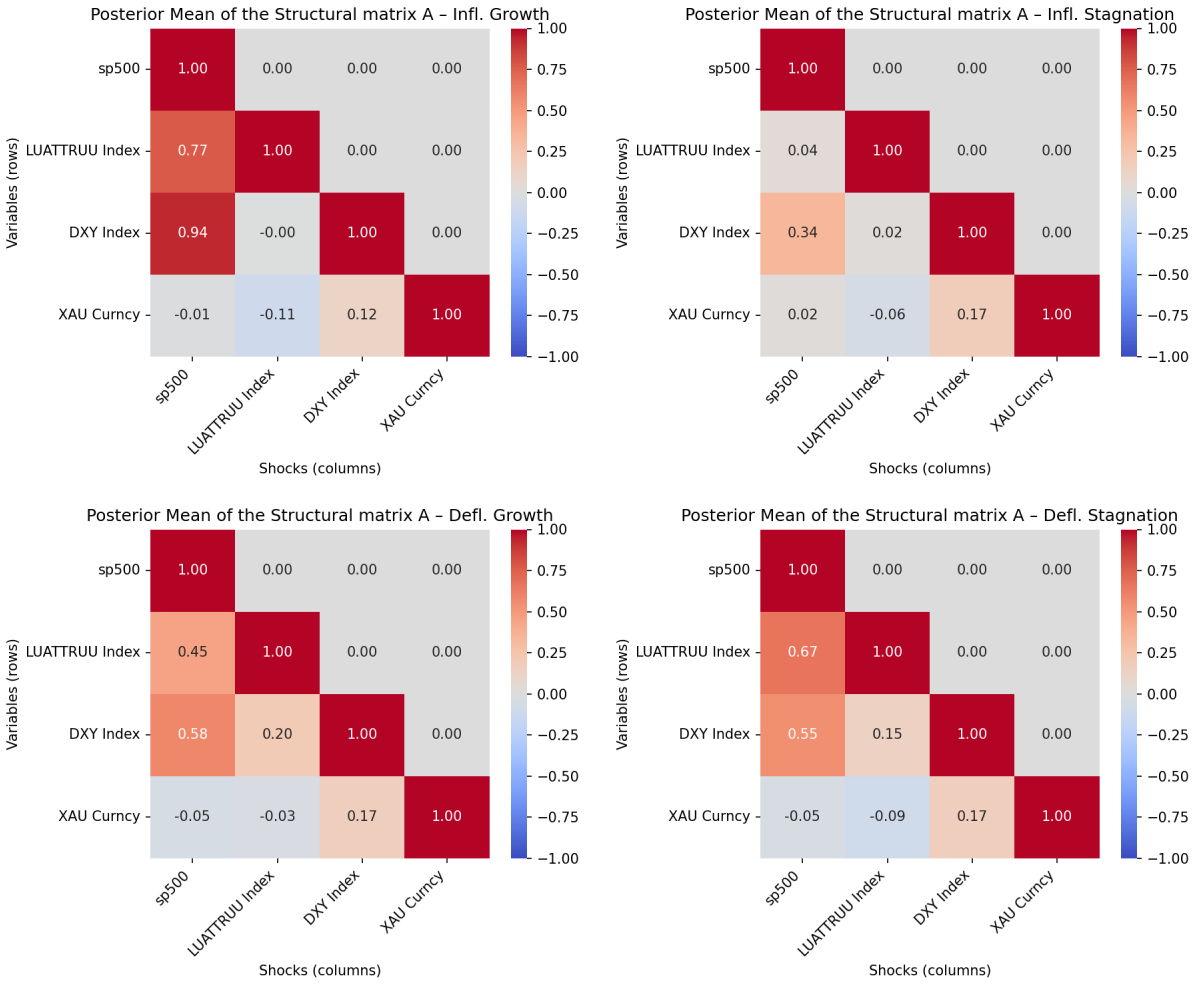


Figure 5.6: Posterior Mean Structural Matrices Across Regimes

gold as a partially segmented safe-haven asset during periods of economic weakness.

5.4.2 Impulse Response Functions

To further analyse the dynamic propagation of shocks, impulse response functions are computed for each regime. The impulse responses trace the effect of a one-unit structural shock on the future path of all asset classes.

For readability, the main text focuses on the impulse responses for the inflationary growth and inflationary stagnation regimes, which provide the strongest contrast between favourable and adverse macro-financial environments. The complete set of impulse response functions for all regimes is reported in the Appendix.

The impulse responses reveal several important features. First, shocks tend to dissipate relatively quickly across all regimes, with most responses converging back toward zero within

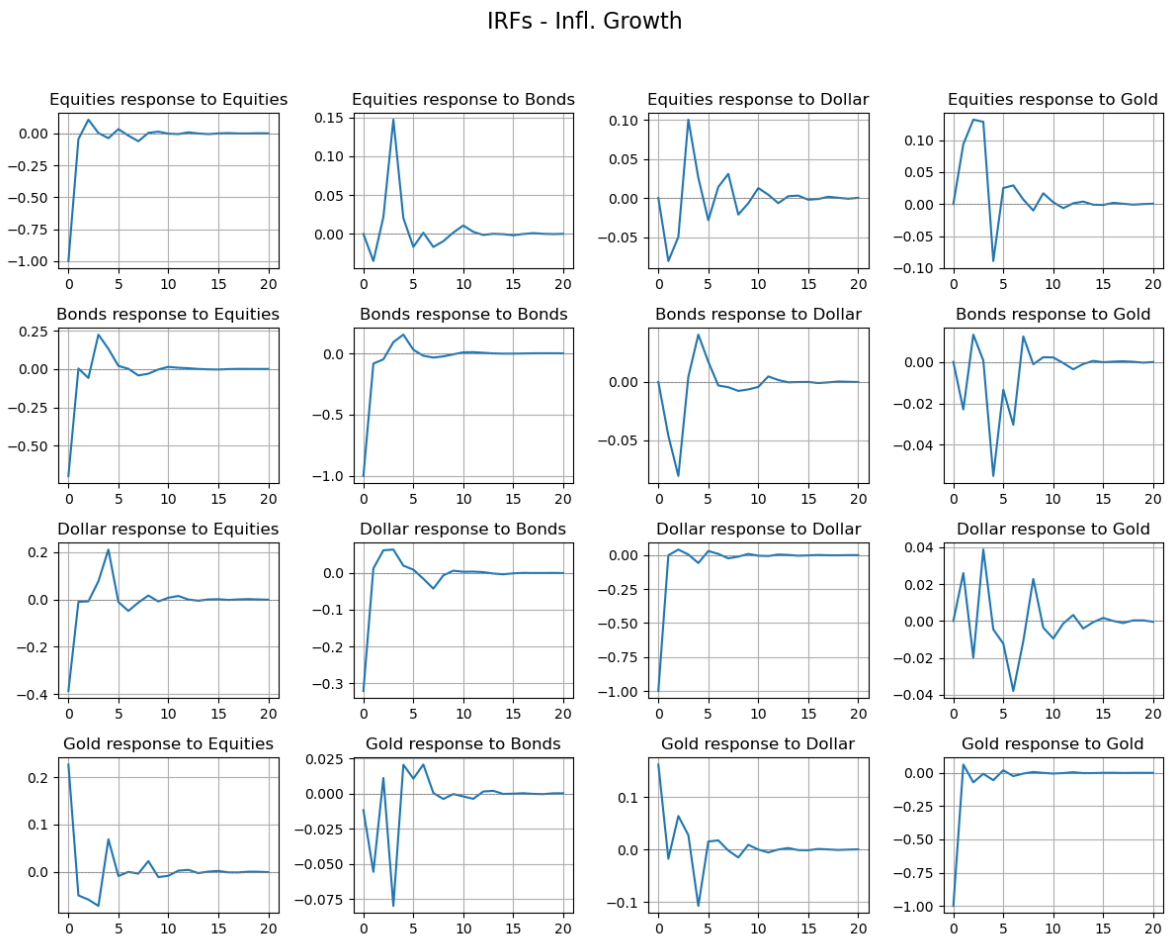


Figure 5.7: Impulse Response Functions – Inflationary Growth

approximately 5 to 10 months. This suggests that the identified shocks primarily capture medium-term financial-market adjustments rather than permanent structural shifts.

Second, the magnitude and propagation of shocks differ considerably across regimes. In the inflationary growth regime, shocks originating from equities generate stronger spillover effects toward bonds and the U.S. dollar. This indicates a highly interconnected financial environment in which movements in risky assets rapidly propagate through the broader financial system.

By contrast, the inflationary stagnation regime shows weaker and more fragmented transmission dynamics. Asset responses are generally smaller and less persistent, suggesting that financial markets behave more defensively and less synchronously during stagflationary environments.

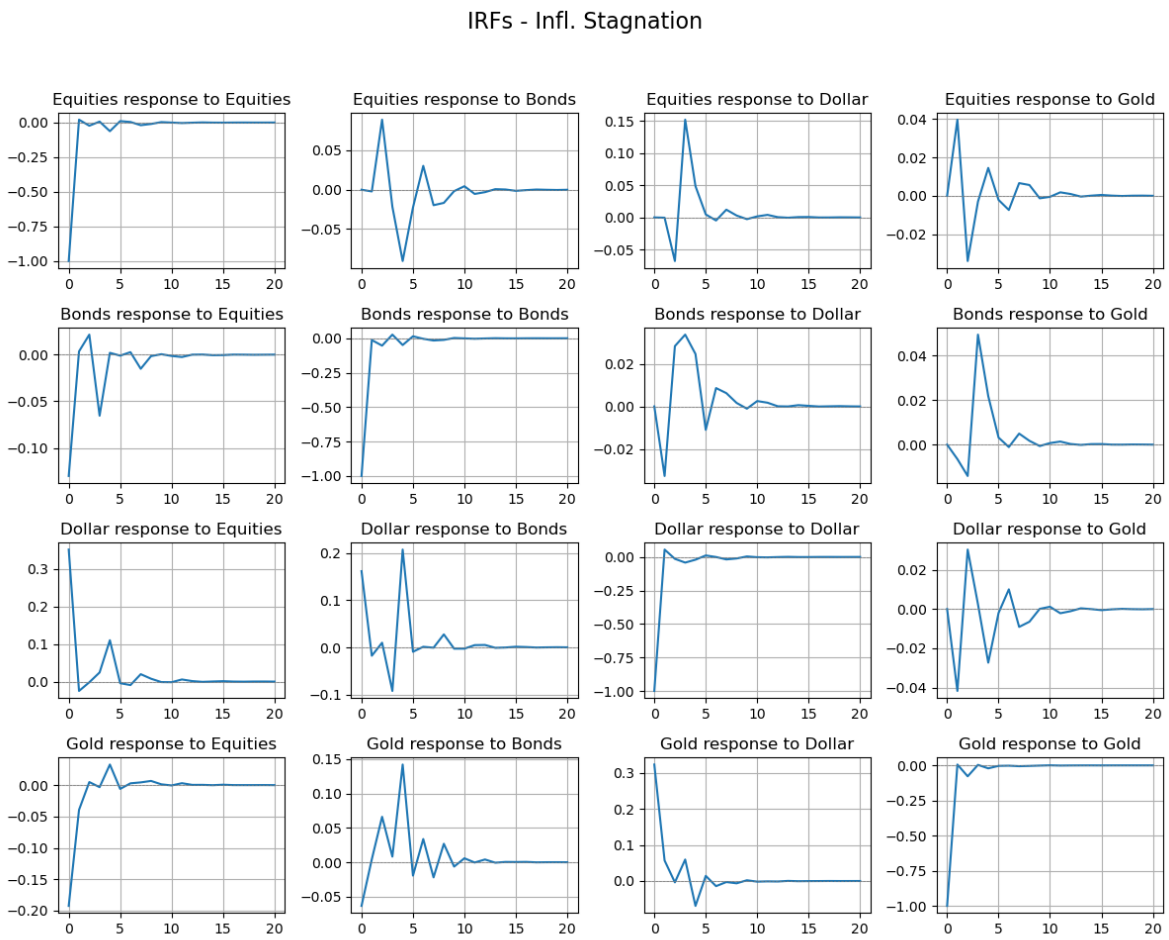


Figure 5.8: Impulse Response Functions – Inflationary Stagnation

5.4.3 Forecast Error Variance Decomposition

The forecast error variance decomposition provides further insight into the relative importance of different structural shocks in explaining forecast uncertainty. Table 5.1 reports the decomposition of the forecast error variance of the S&P 500 at a forecast horizon of 20 months. The decomposition results indicate that the sources of equity-market forecast uncertainty differ substantially across regimes and are generally driven by multiple interacting macro-financial factors rather than a single dominant shock. This highlights the importance of accounting for regime-dependent transmission mechanisms when analyzing cross-asset dynamics.

During inflationary growth, forecast uncertainty is relatively broadly distributed across shocks. Gold shocks account for the largest share of S&P 500 forecast uncertainty at approximately 36%, followed by U.S. dollar shocks at 27% and equity shocks at 26%. Bond shocks

Table 5.1: Forecast Error Variance Decomposition of the S&P 500 at Horizon 20

Regime	Equity Shock	Bond Shock	DXY Shock	Gold Shock
Inflationary Growth	0.26	0.10	0.27	0.36
Inflationary Stagnation	0.45	0.11	0.29	0.15
Deflationary Growth	0.29	0.47	0.14	0.10
Deflationary Stagnation	0.31	0.27	0.22	0.24

contribute only modestly in this regime. This pattern suggests that inflationary expansion phases are characterized by broad macro-financial interactions involving inflation expectations, commodity dynamics, and international capital flows.

Inflationary stagnation is primarily driven by equity-market specific shocks, which explain approximately 45% of forecast uncertainty. At the same time, U.S. dollar shocks remain highly relevant, accounting for roughly 29% of forecast variance. Gold and bond shocks play comparatively smaller roles. This configuration is economically plausible for stagflationary environments, where equity-market uncertainty and currency dynamics are both strongly influenced by inflationary pressure, restrictive monetary policy, and deteriorating growth expectations.

In the deflationary growth regime, bond shocks become the dominant source of forecast uncertainty and explain approximately 47% of the S&P 500 forecast variance. Equity shocks remain important with a contribution of roughly 29%, while dollar and gold shocks are less influential. This result suggests that interest-rate dynamics and fixed-income markets play a particularly important role during stable low-inflation growth periods, where monetary conditions strongly affect equity valuations.

Finally, the deflationary stagnation regime exhibits the most balanced decomposition across shocks. Equity, bond, U.S. dollar, and gold shocks contribute relatively evenly to forecast uncertainty, with shares ranging between approximately 22% and 31%. This pattern is consistent with broad systemic uncertainty during recessionary or crisis-like environments, in which multiple macro-financial channels simultaneously influence asset-market behaviour.

5.5 Bayesian Model Diagnostics

This section evaluates the statistical adequacy of the estimated model using Bayesian model diagnostics. In particular, the analysis focuses on the Marginal Data Density (MDD), the Savage-Dickey Density Ratio (SDDR), and posterior sampling diagnostics. These measures provide insight into model fit, parameter stability, and the overall plausibility of the estimated

posterior distribution.

5.5.1 Marginal Data Density

The Marginal Data Density provides a Bayesian measure of overall model fit by integrating the likelihood over the full prior distribution of the parameter space. Unlike classical likelihood measures, the MDD naturally penalizes model complexity and therefore balances explanatory power against overparameterization [15].
Several estimations of the model using different parameters and lags were conducted in order to assess the numerical stability of the Bayesian estimation procedure. Across these estimation runs, the resulting MDD values differed moderately, which is not uncommon in high-dimensional Bayesian regime-switching models estimated using simulation-based methods. Since the MDD is approximated using importance sampling, its numerical value can be sensitive to posterior sampling variability and the construction of the proposal distribution. Among the estimated specifications, the most economically and statistically plausible model produced a log marginal data density of

$$\log p(Y) = 308.83. \tag{5.2}$$

Compared to the alternative estimation runs, this specification simultaneously exhibited the most stable regime classification, economically interpretable regime dynamics, and coherent posterior behaviour. Consequently, the model introduced in Chapter 5.1 is used as the baseline specification throughout the remainder of the empirical analysis.
The positive value of the log MDD indicates substantial empirical support for the estimated specification relative to alternative parameter configurations.

5.5.2 Savage-Dickey Density Ratio

In addition to overall model fit, Bayesian evidence for parameter restrictions is evaluated using the Savage-Dickey Density Ratio (SDDR). The SDDR compares posterior and prior densities at restricted parameter values and therefore provides a Bayesian measure of evidence for structural relationships within the model.
Table 5.2 reports the estimated log-scale SDDR values for the baseline specification. The reported SDDR values indicate substantial posterior evidence for differences across regimes. In particular, the relatively large values between several regime combinations suggest that the posterior distributions differ meaningfully from the corresponding prior distributions. This further shows that the identified regimes capture genuinely distinct financial-market

Table 5.2: Savage-Dickey Density Ratio (log-scale)

	Regime 1	Regime 2	Regime 3	Regime 4
Inflationary Growth	–	3.69	3.18	3.71
Inflationary Stagnation	3.69	–	2.22	1.65
Deflationary Growth	3.18	2.22	–	3.87
Deflationary Stagnation	3.71	1.65	3.87	–

environments rather than reflecting purely statistical noise.

The strongest differences emerge between Deflationary Growth and Deflationary Stagnation, as well as between Inflationary Growth and Deflationary Stagnation. Economically, this is consistent with the interpretation that growth and stagnation regimes exhibit fundamentally different cross-asset return and volatility dynamics.

5.5.3 Posterior Sampling Diagnostics

Additional posterior diagnostics are reported in the Appendix. Figure A.4 presents trace plots and posterior density estimates for the main parameter blocks of the Gibbs sampler, including VAR coefficients, structural parameters, transition probabilities, variance parameters, and hyperparameters.

Overall, the posterior draws fluctuate around relatively stable posterior means without exhibiting explosive behaviour or systematic trends. Most parameter chains show satisfactory mixing behaviour, particularly for the VAR coefficients and transition probabilities. Some persistence and occasional spikes remain visible for a subset of hyperparameters and structural parameters, which is not uncommon in high-dimensional Bayesian models. Nevertheless, the posterior distributions remain economically plausible and stable across the retained draws.

5.6 Out-of-Sample Forecast Evaluation

In this section the predictive performance of the model is evaluated using an out-of-sample forecasting exercise over the final 36 months of the sample period. Forecasts are generated recursively using only information available at the respective forecast date, thereby avoiding look-ahead bias and ensuring a realistic real-time evaluation framework.

Since forecasting the exact magnitude of financial returns is inherently difficult, the evaluation focuses primarily on the directional accuracy of forecasts. For each asset class, the model generates a binary prediction indicating whether the subsequent monthly return is

expected to be positive or negative. The forecasting performance is then assessed using standard binary classification measures, including accuracy, precision, recall, and the F1-score. Figures 5.9 – 5.12 compare the realized sign of returns with the predicted sign generated by the model. In addition, the figures report the estimated probability of a positive return derived from the posterior predictive distribution.

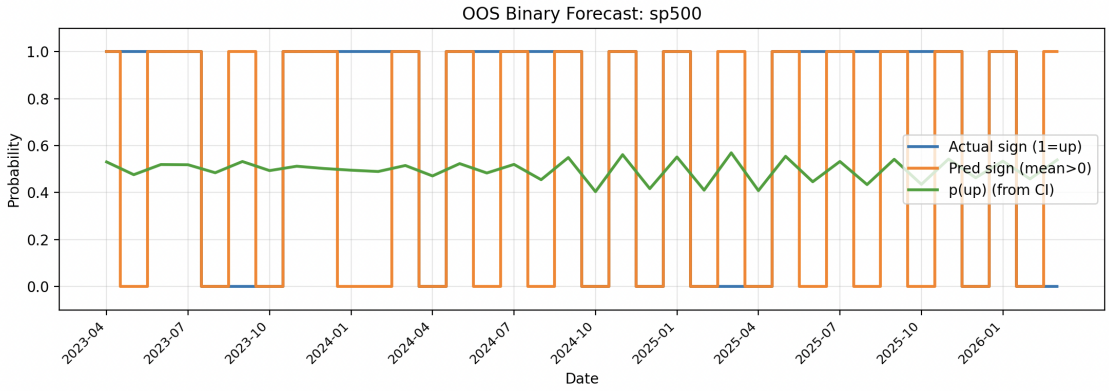


Figure 5.9: Out-of-Sample Binary Forecast – S&P 500

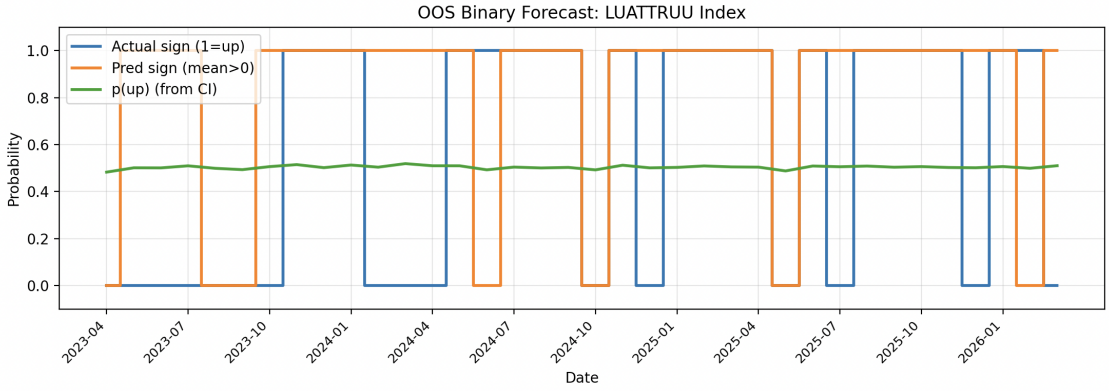


Figure 5.10: Out-of-Sample Binary Forecast – Bonds

Table 5.3 summarizes the out-of-sample forecast performance across asset classes. The positive class is defined as a positive monthly return. True positives (TP) therefore correspond to correctly predicted positive return months, while true negatives (TN) indicate correctly predicted negative return months. False positives (FP) occur when the model predicts a positive return despite a realized negative return, whereas false negatives (FN) indicate missed positive return periods.

Overall, the results suggest that the model contains moderate but economically meaningful predictive information. Directional accuracy ranges between approximately 63.9% and 75.0% across the four asset classes. Although forecasting financial returns remains difficult,

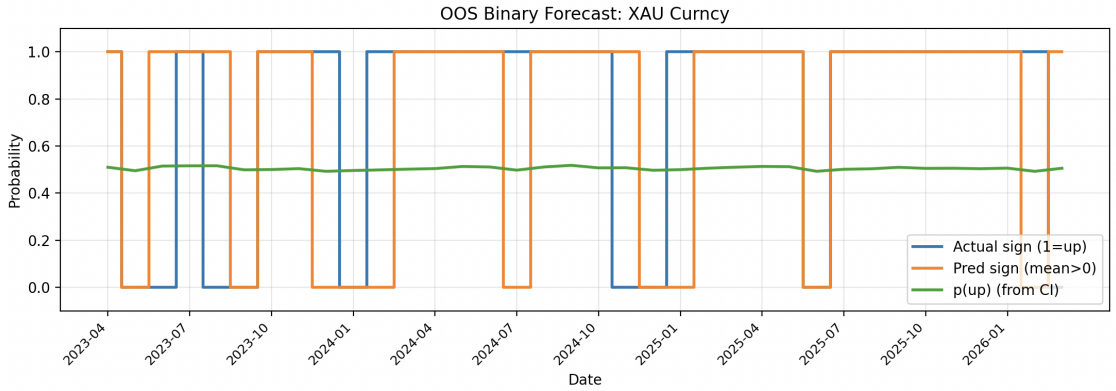


Figure 5.11: Out-of-Sample Binary Forecast – Gold

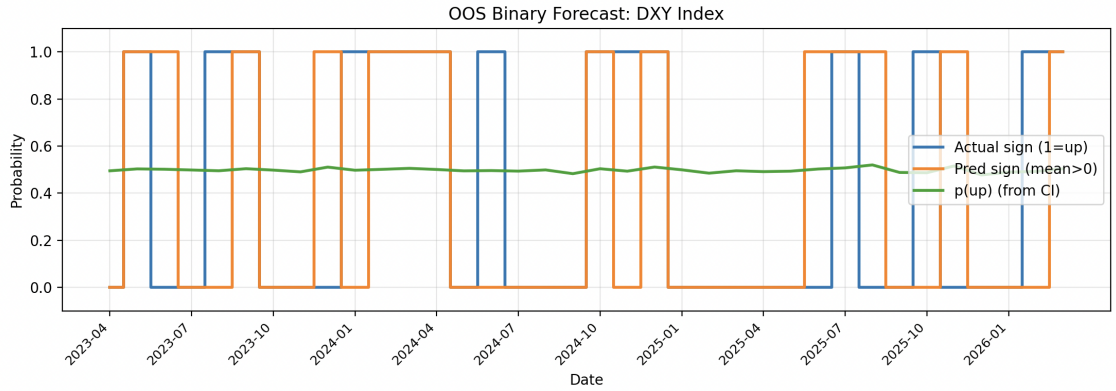


Figure 5.12: Out-of-Sample Binary Forecast – U.S. Dollar Index

the results indicate that the regime-dependent framework captures stable directional tendencies across multiple asset classes over the out-of-sample evaluation period.

The strongest overall forecasting performance is observed for gold, which achieves an accuracy of 75.0% together with comparatively strong precision and recall values. The S&P 500 and the U.S. Dollar Index both achieve directional accuracies of approximately 69.4%, suggesting that the framework captures useful directional information for both risky assets and currency markets.

The bond forecasts exhibit a comparatively balanced performance. The model correctly identifies eighteen positive and only five negative bond-return months, while generating an elevated number of false positives. The resulting precision of 62.1% indicates the worst predicting power of the asset classes and shows that not all positive predicted positives are indeed realized. Nevertheless, the recall of 90.0% shows that the large majority of positive bond periods are captured.

For the U.S. Dollar Index, the model achieves balanced forecasting performance with an accuracy of approximately 69.4%. The results suggest that the framework captures directional

Table 5.3: Out-of-Sample Binary Forecast Performance

Asset	TP	FP	TN	FN	Accuracy	Precision	Recall	F1
S&P 500	16	3	9	8	0.694	0.842	0.667	0.742
Bonds	18	11	5	2	0.639	0.621	0.900	0.735
DXY	9	5	16	6	0.694	0.643	0.600	0.621
Gold	22	4	5	5	0.750	0.846	0.815	0.830

movements in currency markets reasonably well, although forecasting performance remains more moderate compared to gold and equities. The precision of approximately 64.3% indicates that a substantial share of predicted positive dollar movements are realized correctly, while the recall of 60.0% suggests that some positive return periods remain difficult to identify in real time.

For gold, the forecasting results remain comparatively strong throughout the evaluation period. The model captures both positive and negative return phases relatively consistently, which is reflected in the comparatively high precision and recall values reported in Table 5.3. This behaviour is economically plausible, as gold often reacts more systematically to shifts in uncertainty, inflation expectations, and defensive market positioning than more cyclical asset classes. More generally, the forecast probabilities displayed in Figures 5.9 – 5.12 remain relatively close to 0.5 throughout the evaluation period. This is economically plausible, as expected excess returns are generally small and financial markets remain difficult to predict. Importantly, the model does not produce unrealistically extreme forecast probabilities, which suggests that the posterior predictive distribution appropriately reflects the substantial uncertainty surrounding future financial-market movements.

5.7 Portfolio Backtest Results

The final step of the empirical analysis evaluates whether the regime-dependent forecasts generated by the model can be translated into economically meaningful investment strategies. To assess the practical relevance of the forecasting framework, the out-of-sample predictions are converted into trading signals and evaluated within a portfolio backtesting exercise.

Two different portfolio strategies are considered. The first strategy is a long-only forecast portfolio, which allocates equal weight to all assets with a positive forecast signal. The second strategy is a long-short portfolio, which takes long positions in assets with positive expected returns and short positions in assets with negative expected returns. Both strategies are evaluated including transaction costs when reweighting and are compared against two benchmark portfolios: an equal-weight buy-and-hold portfolio and a passive S&P 500

investment.

5.7.1 Portfolio Performance

Figure 5.13 reports the cumulative portfolio performance over the three-year out-of-sample evaluation period. Several important observations emerge from the results. First, all forecast-

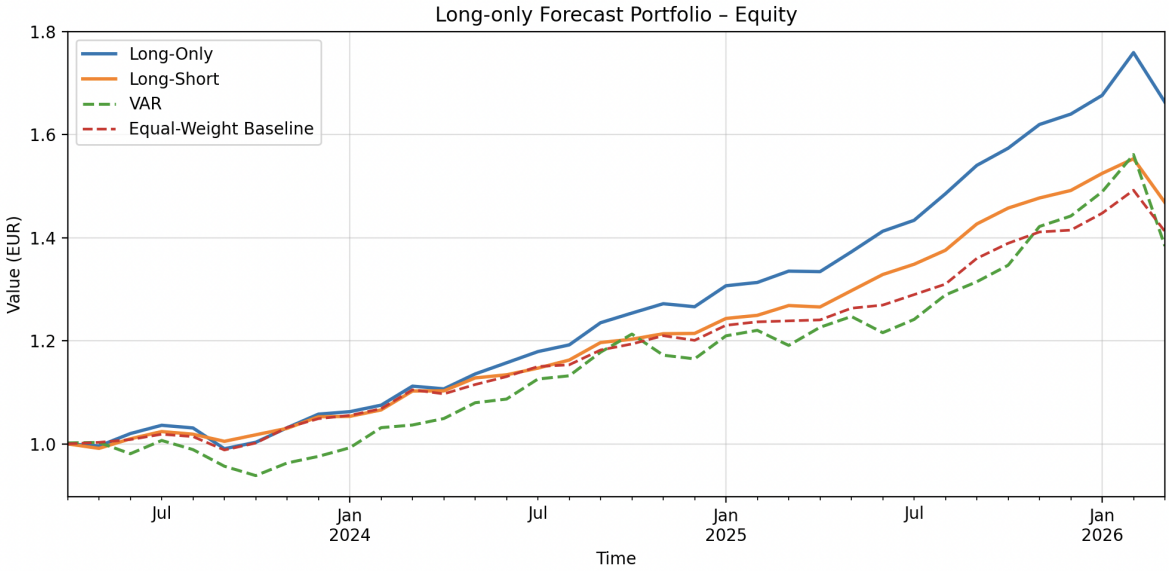


Figure 5.13: Cumulative Portfolio Performance

driven strategies generate positive cumulative returns over the evaluation period. The long-only strategy performs particularly strongly and outperforms both the equal-weight benchmark and the VAR benchmark during most of the sample period.

The long-only strategy finishes the evaluation period with a cumulative gain of approximately 66.3%, corresponding to an annualized return of approximately 22.1%. At the same time, the strategy exhibits comparatively moderate annualized volatility of around 7.0% and a maximum drawdown of approximately -5.5%. This suggests that the regime-dependent forecasting framework generates comparatively stable portfolio allocations while maintaining attractive risk-adjusted performance.

The long-short strategy also produces positive returns, although performance remains weaker compared to the long-only implementation. The cumulative gain amounts to approximately 46.9%, while annualized volatility reaches roughly 5.6%. In addition, the maximum drawdown remains close to -5.5%. This indicates that the inclusion of short positions does not necessarily improve portfolio performance and may reduce the ability of the framework to exploit directional market movements effectively. Thus, the inclusion of negative expectations seems to slightly reduce volatility at the cost of higher returns.

The VAR benchmark also generates positive returns over the evaluation period, reaching a cumulative gain of approximately 38.2%. However, the strategy exhibits noticeably higher annualized volatility of around 11.0% together with a substantially larger maximum drawdown of approximately -11.4%. Consequently, the regime-dependent framework appears to provide more favourable risk-adjusted portfolio dynamics relative to a standard linear VAR specification without latent regime structure.

Table 5.4: Portfolio Backtest Performance

Strategy	CAGR	Annualized Volatility	Max Drawdown	Total Gain
Long-Only Forecast	0.221	0.070	-0.055	0.663
Long-Short Forecast	0.156	0.056	-0.055	0.469
VAR Benchmark	0.127	0.110	-0.114	0.382

Interestingly, the long-only strategy also compares favourably against the benchmark portfolios shown in Figure 5.13. Although the equal-weight benchmark and the VAR benchmark occasionally approach the performance of the forecast-driven strategies during certain subperiods, the long-only implementation remains the strongest overall performer for most of the evaluation window. The difference in performance seems to emerge at around one year into the out of sample data. In addition, the long-only strategy exhibits comparatively smoother portfolio dynamics and more moderate drawdowns than the VAR benchmark.
Overall, the results suggest that the regime-dependent forecasting framework contains economically useful information that can be translated into systematic portfolio allocation decisions. Importantly, the superior performance of the long-only strategy appears to stem not only from higher cumulative returns, but also from its comparatively favourable risk-adjusted characteristics. Nevertheless, the out-of-sample evaluation period remains limited from a statistical perspective, and the reported performance should therefore not be interpreted as definitive evidence of persistent market predictability.

5.7.2 Comparison with Random Buy-and-Hold Portfolios

To further evaluate the economic significance of the forecast-based strategies, the portfolio results are compared against a large set of randomly generated buy-and-hold portfolios. Specifically, 1,000 random portfolios are simulated by drawing static portfolio weights from a Dirichlet distribution and holding these weights constant throughout the evaluation period. Figure 5.14 presents the resulting distribution of terminal portfolio values together with the final values of the long-only and long-short forecast strategies.

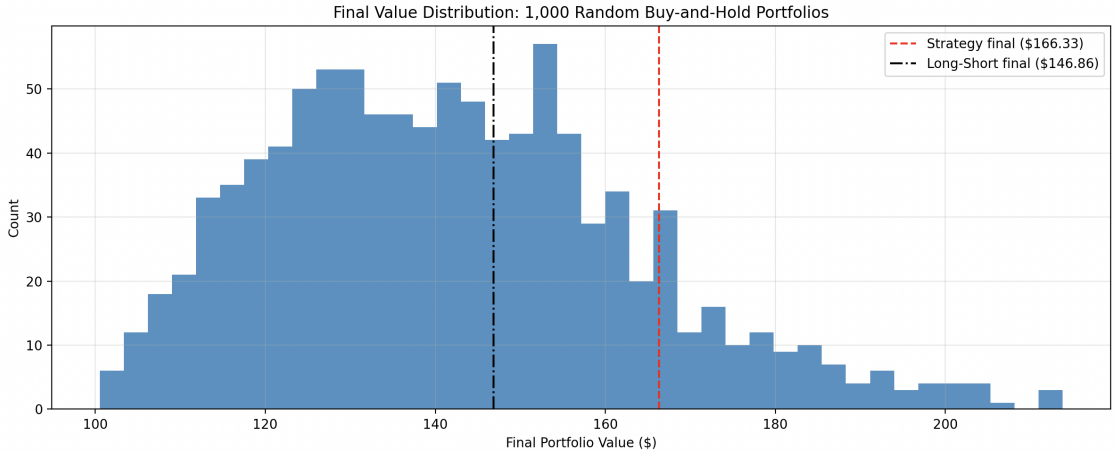


Figure 5.14: Distribution of Final Values for 1,000 Random Buy-and-Hold Portfolios

The results indicate that the long-only forecast strategy lies in the upper region of the random portfolio distribution. The final value of the long-only strategy exceeds a substantial share of randomly generated buy-and-hold allocations, suggesting that the forecasting framework provides information beyond naive static diversification.

The long-short strategy also performs above the centre of the random portfolio distribution, although its performance remains noticeably below the long-only strategy. This again indicates that the additional complexity introduced by short positions does not necessarily improve portfolio performance during the evaluation period.

Importantly, the comparison against random portfolios provides an economically intuitive robustness check. Since the random portfolios represent a broad range of passive allocation strategies, outperforming a large share of these portfolios suggests that the model-based forecasts contain relevant information rather than merely benefiting from favourable market exposure. Moreover, the primary advantage of the long-only implementation lies in its ability to generate comparatively attractive returns while maintaining moderate drawdowns and volatility.

5.8 Robustness and Limitations

While the empirical results provide evidence that the proposed framework captures regime-dependent dynamics, several limitations should be acknowledged. These limitations are important for interpreting the results cautiously and for assessing the robustness of the empirical findings.

First, the model is relatively complex compared to the available sample size. The framework includes multiple latent regimes, regime-specific VAR coefficients, regime-dependent covari-

ance structures, structural matrices, transition probabilities, and latent regime paths. This flexibility is useful because it allows the model to capture changing market environments, but it also increases the risk of overparameterization and parameter instability. Although Bayesian shrinkage priors and symmetric prior specifications are used to stabilize estimation, some regime-specific parameters may still be sensitive to the chosen prior structure, initialization values, and sample composition.

Second, the regime interpretation remains partly model-based. The regimes are labelled as inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation, but the model itself is estimated only on financial asset returns. It does not directly include observable macroeconomic variables such as inflation, policy rates, industrial production, credit spreads, or unemployment. Therefore, the macro-financial interpretation of the regimes should be understood as an economic interpretation of latent return patterns rather than as a direct classification of observed macroeconomic states. Including explicit macroeconomic indicators could strengthen the link between the estimated regimes and observable economic conditions.

Third, although the out-of-sample evaluation window contains three years, the forecasting exercise remains limited from a statistical perspective. Financial return predictability is difficult to evaluate over short samples, and portfolio performance may be influenced by a small number of favourable or unfavourable market episodes. Consequently, the reported forecasting accuracy and backtest results should not be interpreted as definitive evidence of persistent market predictability. Instead, they suggest that the regime-dependent framework contains useful directional information during the considered evaluation period.

Fourth, the benchmark analysis improves the empirical robustness of the thesis, but it remains limited. The proposed framework is compared against passive benchmark portfolios, random buy-and-hold allocations, and a standard VAR benchmark. These comparisons indicate that the regime-dependent model provides added value relative to simpler alternatives. Nevertheless, further benchmark comparisons could strengthen the analysis. In particular, future work could compare the model against simpler Markov-switching models, Bayesian VAR specifications without structural regimes, or financial trading rules based on momentum, trend-following, and volatility timing.

Finally, the estimation of high-dimensional Bayesian regime-switching models is computationally demanding. The Gibbs sampler with embedded Metropolis-Hastings steps requires careful initialization, diagnostic evaluation, and numerical stabilization. While the posterior diagnostics indicate broadly stable sampling behaviour, some persistence in selected parameter chains remains visible. This is not unusual in complex Bayesian models, but it reinforces the need to interpret individual parameter estimates with caution and to focus primarily on

the broader economic patterns that are stable across the empirical analysis.

Overall, these limitations do not invalidate the main findings of the thesis, but they provide important context for their interpretation. The results should be viewed as evidence that regime-dependent Bayesian structural models can offer a useful and interpretable framework for analysing cross-asset dynamics, rather than as final proof of stable and persistent forecasting superiority. Future research could build on this framework by incorporating observable macroeconomic variables, testing alternative regime specifications, extending the out-of-sample evaluation, and comparing the model against a broader set of forecasting and portfolio benchmarks.

CHAPTER 6: CONCLUSION

This thesis examined whether regime-dependent dynamics in financial markets can be captured using a Markov-Switching Heteroskedastic Bayesian Structural Vector Autoregressive model and whether the resulting regime information can be used for economic interpretation and portfolio decision-making. The analysis was based on monthly returns of four major asset classes: equities, bonds, the U.S. dollar, and gold. By combining Bayesian estimation, Markov-switching dynamics, structural identification, and out-of-sample forecasting, the thesis aimed to provide a coherent framework for understanding cross-asset behaviour across different market environments.

The empirical results show that financial market dynamics are strongly regime dependent. The model identifies four economically interpretable regimes, labelled as inflationary growth, inflationary stagnation, deflationary growth, and deflationary stagnation. These regimes differ meaningfully in their conditional mean returns, volatility patterns, transition behaviour, and structural shock transmission. In particular, the interaction between equities and bonds provides a useful basis for distinguishing growth and stagnation environments, while the behaviour of gold and the U.S. dollar adds further information about inflation pressure, safe-haven demand, and macroeconomic uncertainty.

The conditional mean results indicate that inflationary growth is associated with strong equity performance and weak bond returns, while inflationary stagnation is characterized by simultaneous weakness in equities and bonds together with strong U.S. dollar performance. Deflationary growth provides the most favourable environment for diversified portfolios, as both equities and bonds perform above average. Deflationary stagnation, in contrast, is marked by weak equity returns and strong performance of defensive assets such as bonds and gold. These findings support the central premise of the thesis that asset returns cannot be adequately described by a single stable return-generating process.

The volatility analysis further confirms the relevance of the heteroskedastic specification. Volatility differs substantially across regimes and across asset classes. Importantly, adverse return regimes do not always coincide with the highest volatility, which highlights the value of distinguishing between return conditions and risk conditions. The regime probability estimates also provide economically plausible historical classifications, with crisis periods being associated with defensive or stagnation regimes. The most recent observations are classified as inflationary stagnation, which is consistent with an environment of weaker growth expectations, persistent inflationary pressure, restrictive monetary policy, and elevated geopolitical uncertainty.

The transition matrix provides additional evidence that the estimated regimes represent persistent market environments rather than random short-term fluctuations. The diagonal elements are clearly dominant, implying that regimes tend to last for several months. Inflationary stagnation is the most persistent regime, which is economically plausible given that inflationary pressure and restrictive monetary policy often resolve only gradually.

The structural transmission results show that contemporaneous relationships and shock propagation mechanisms differ across regimes. The structural matrices reveal that cross-asset transmission is strongest in inflationary growth and weaker in inflationary stagnation, indicating that market interconnections change with the macro-financial environment. The impulse response functions show that shocks generally dissipate within a few months, but their magnitude and transmission path differ across regimes. The forecast error variance decomposition further demonstrates that the drivers of equity-market uncertainty are regime specific. Depending on the state, equity uncertainty is explained by a different combination of equity, bond, currency, and gold shocks. This confirms that regime-dependent modelling provides information that would be lost in a constant-parameter framework.

The out-of-sample forecasting exercise provides further support for the economic usefulness of the model. Although predicting financial returns is inherently difficult, the model achieves directional accuracies between 63.9% and 75.0% across the four asset classes during the evaluation window. The results are particularly strong for the S&P 500 and the U.S. Dollar Index. The forecast probabilities remain close to 0.5, which indicates that the model does not generate unrealistically confident predictions, but still contains useful directional information. This is an important result, as even moderate predictive information can be valuable when translated into systematic portfolio rules.

The portfolio backtest shows that the regime-dependent forecasts can be transformed into economically meaningful investment strategies. The long-only forecast portfolio performs particularly strongly and remains the most successful implementation throughout both evaluation horizons. Over the primary one-year out-of-sample period, the strategy generates a cumulative gain of approximately 23.8% while maintaining comparatively low volatility and a very limited drawdown. The extended three-year evaluation further reinforces this result, with the long-only strategy achieving a cumulative gain of approximately 70.3% and continuing to outperform both the equal-weight benchmark and the long-short implementation.

The long-short strategy also produces positive returns across both evaluation windows, but its performance remains noticeably weaker than the long-only approach. This suggests that the forecasting framework is more effective at identifying attractive long exposures than at generating profitable short positions.

The comparison with 1,000 random buy-and-hold portfolios further supports the economic

relevance of the forecasting framework. In both the one-year and three-year evaluations, the long-only forecast portfolio lies clearly in the upper tail of the simulated distribution, indicating that the model generates information beyond naive static diversification. These findings suggest that the proposed framework does not merely provide statistical regime classifications but can also support practical and economically meaningful asset allocation decisions.

Overall, the results of this thesis demonstrate that a Markov-Switching Heteroskedastic Bayesian SVAR is a useful framework for modelling cross-asset dynamics. The model captures nonlinearities, time-varying volatility, regime persistence, and state-dependent shock transmission in a coherent Bayesian setting. The identified regimes are economically interpretable and align with major historical market episodes. Moreover, the regime-dependent forecasts contain information that can be translated into portfolio strategies with favourable risk-return characteristics.

Nevertheless, several limitations remain. First, the model is computationally demanding, and the Bayesian estimation procedure requires careful initialization, prior selection, and diagnostic evaluation. Second, the regime labels are based on financial-market behaviour rather than directly observed macroeconomic variables, which means that the economic interpretation remains partly model-based. Finally, the estimation of the marginal data density can be sensitive to simulation uncertainty in high-dimensional regime-switching models.

Future research could extend the analysis in several ways. A natural extension would be to include explicit macroeconomic variables such as inflation, industrial production, policy rates, or credit spreads in order to strengthen the link between latent regimes and observable economic conditions. Further research could also compare the four-regime specification with alternative regime numbers or time-varying transition probabilities. Finally, alternative portfolio construction methods, such as volatility targeting or mean-variance optimization, could be used to assess whether the predictive information contained in the regimes can be exploited even more effectively.

In conclusion, the thesis shows that regime-switching Bayesian structural models provide a powerful and interpretable tool for analysing financial markets. By allowing return dynamics, volatility structures, and shock transmission mechanisms to change across latent states, the model captures essential features of real-world financial behaviour. The empirical findings support the view that market regimes matter not only for explaining historical asset performance, but also for forecasting and portfolio allocation.

APPENDIX A: ADDITIONAL FIGURES

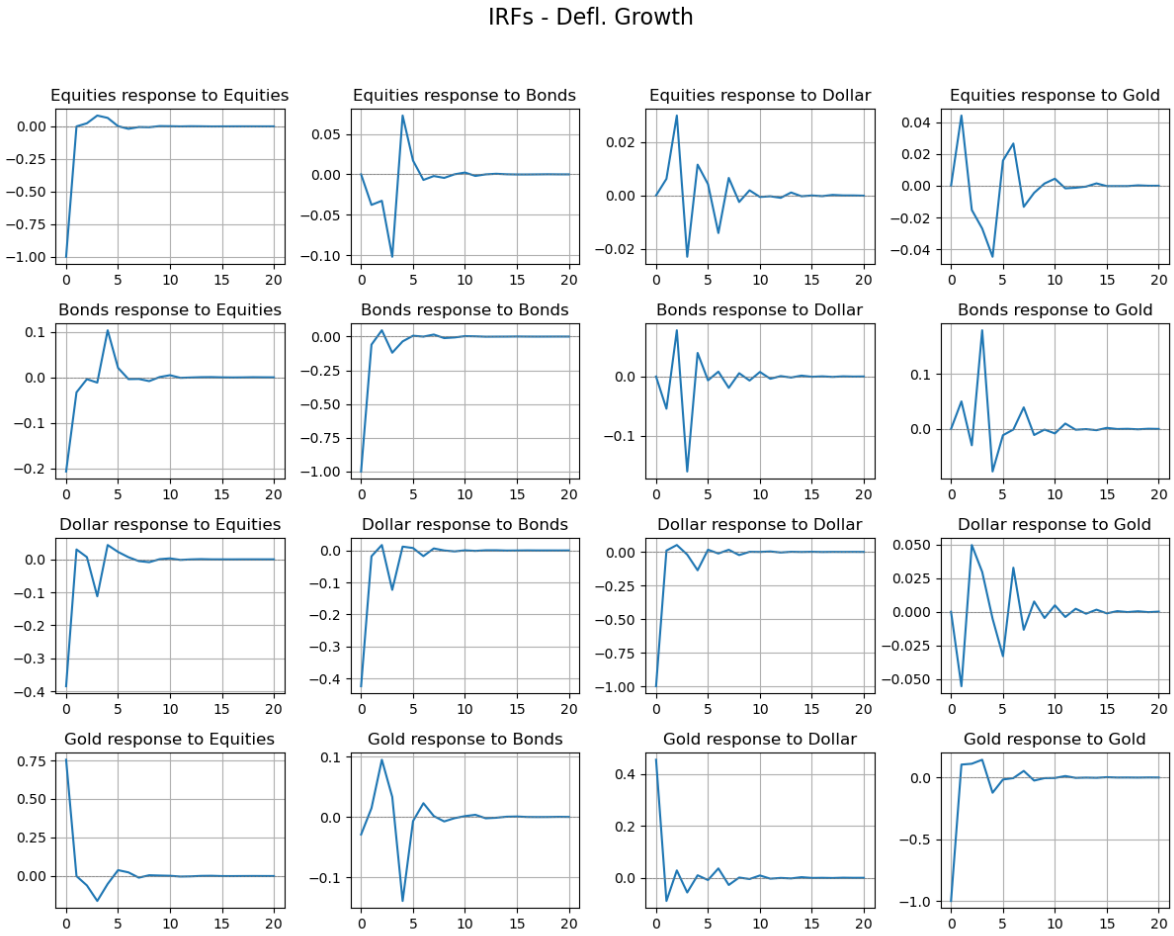


Figure A.1: Impulse Response Functions – Inflationary Growth

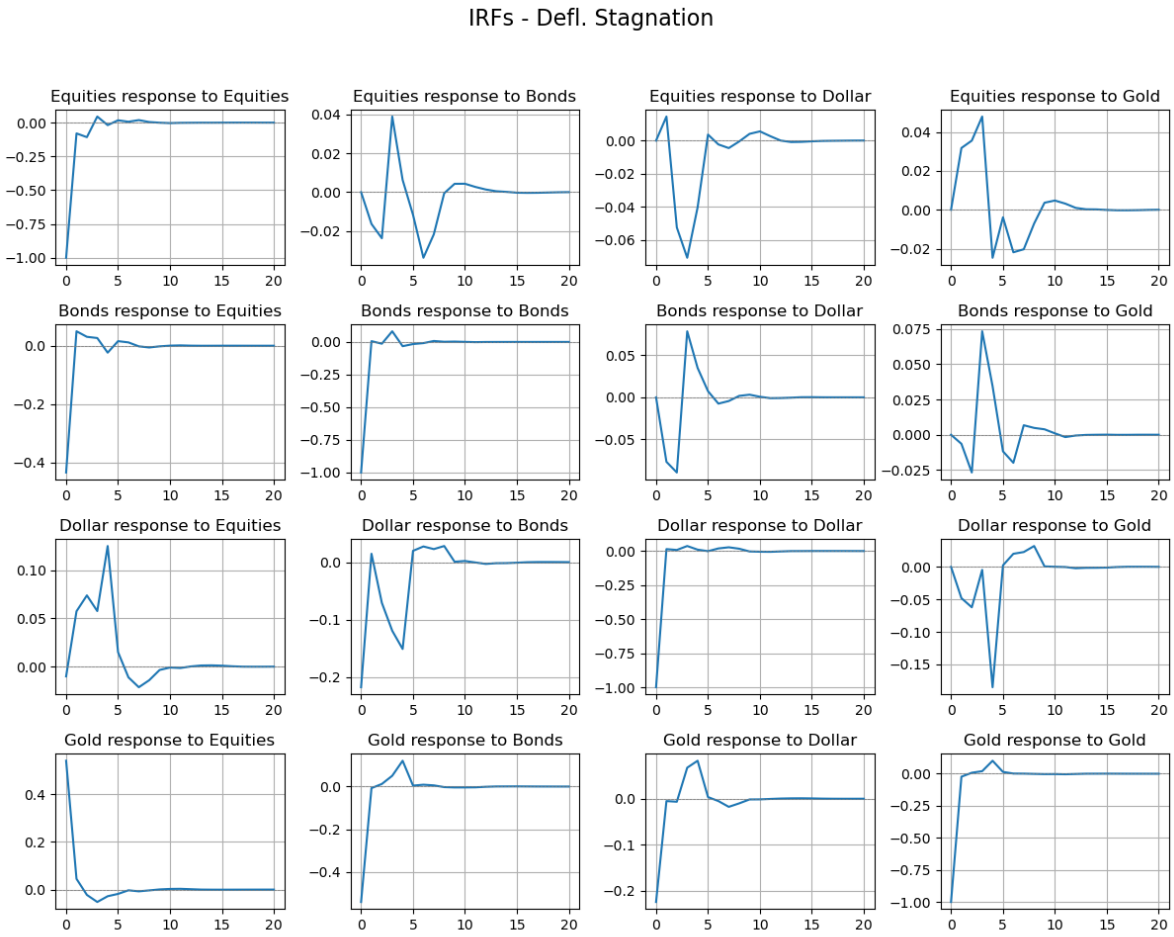


Figure A.2: Impulse Response Functions – Inflationary Stagnation

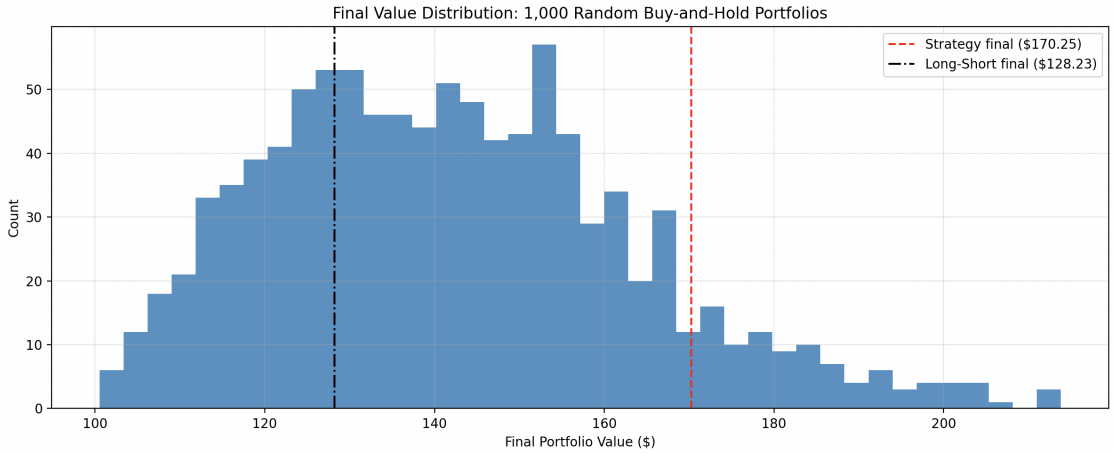
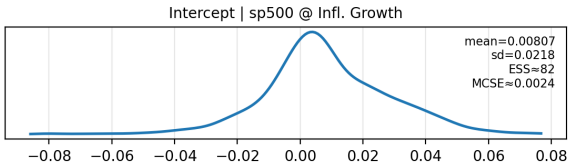
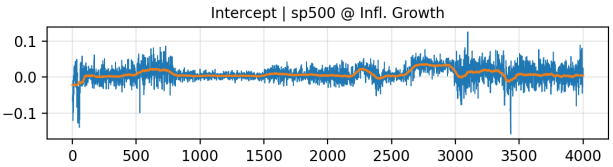
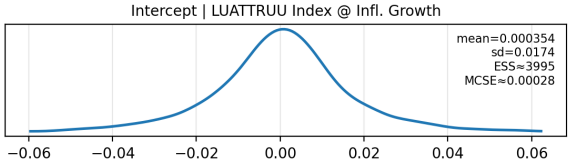
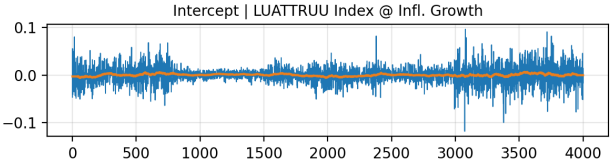


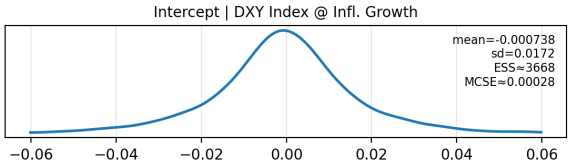
Figure A.3: Distribution of Final Values for 1,000 Random Buy-and-Hold Portfolios: Three-Year Extension

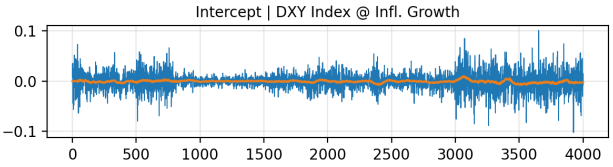


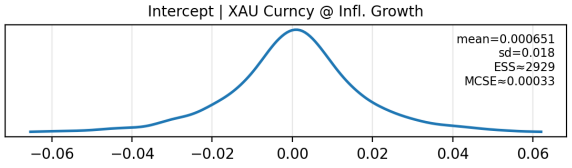


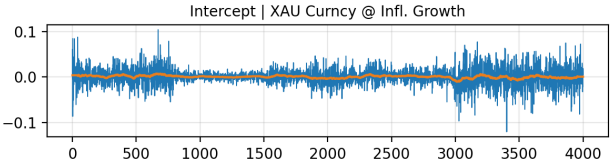


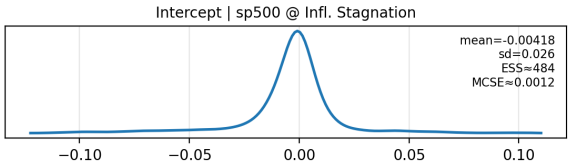


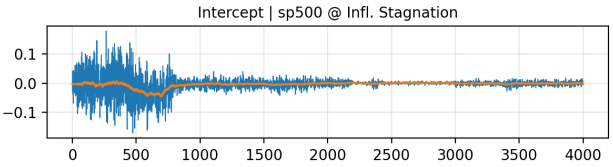


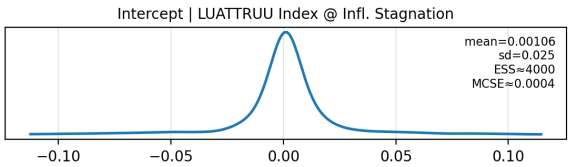


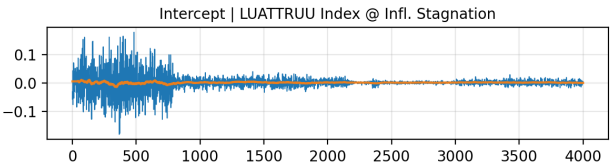


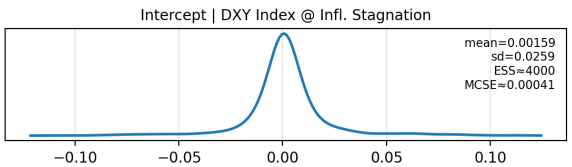


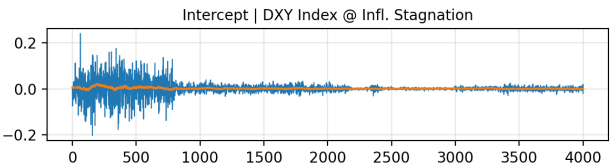


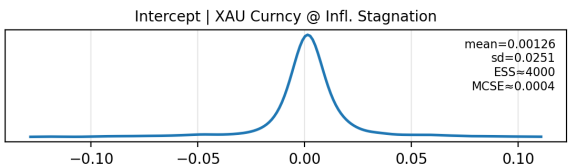


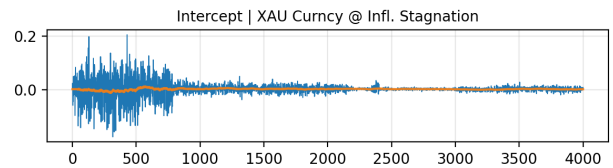


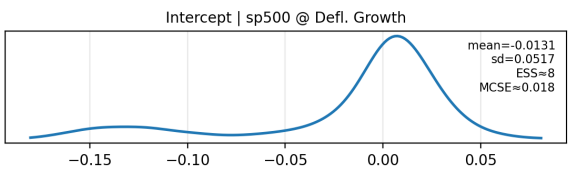


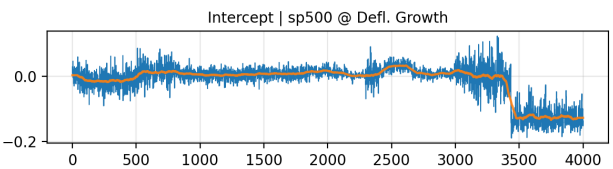


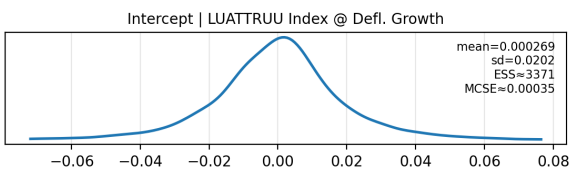


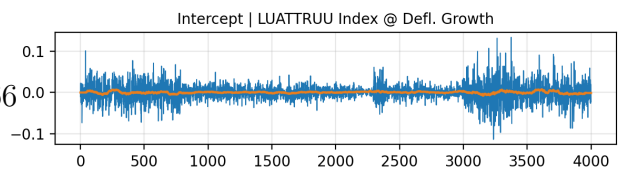












Intercept | DXY Index @ Defl. Growth

Intercept | DXY Index @ Defl. Growth

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